

Computer Vision (EE6102)

Frequency Domain Processing

2026 Fall

Review : Signal as a Linear Combination of Basis Functions

- Vector-space view of a signal
 - In linear algebra, any vector can be represented as a linear combination of basis vectors

$$\mathbf{x} = a_1 \mathbf{b}_1 + a_2 \mathbf{b}_2 + \cdots + a_N \mathbf{b}_N$$

- $\mathbf{b}_1, \dots, \mathbf{b}_N$: basis vector, must be linear independent (Orthogonality is not required)
 - $a_1, \dots, a_N$: coefficients
- Example

$$\mathbf{x} = a_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Review : Signal as a Linear Combination of Basis Functions

- A signal can also be viewed as a vector
- Instead of ordinary basis vectors, we use **basis functions**

$$x(t) = \sum_k a_k \phi_k(t)$$

- $\phi_k(t)$: basis functions
- a_k : contribution of each basis function
- The choice of basis depends on what property of the signal we want to describe.
- For frequency analysis, a natural choice is a set of sinusoidal functions with different frequencies

$$\cos(\omega t), \sin(\omega t)$$

A general sinusoidal components is written as

$$A \cos(\omega t + \phi) \rightarrow A e^{j(\omega t + \phi)} = A e^{j\phi} e^{j\omega t}$$

Review : Signal as a Linear Combination of Basis Functions

For Fourier analysis, the basis functions are complex sinusoidals

$$\phi_w(t) = e^{j\omega t}$$

Or equivalently,

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$$

Thus, a signal is represented as a combination of sinusoidal components with different frequencies .

Consequently

$$x(t) = \text{combination of } e^{j\omega t} \text{ at different frequencies}$$

The Fourier transform tell us how much of each frequency basis is presented in the signal

Review : From Vector Projection to Fourier Transform

For an orthogonal basis, a coefficient is obtained by projection:

$$a_k = \langle \mathbf{x}, \mathbf{b}_k \rangle.$$

For continuous signals, the inner product becomes

$$\langle x(t), \phi_\omega(t) \rangle = \int_{-\infty}^{\infty} x(t) \phi_\omega^*(t) dt.$$

Since

$$\phi_\omega(t) = e^{j\omega t},$$

its complex conjugate is

$$\phi_\omega^*(t) = e^{-j\omega t}.$$

Therefore,

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

which is the Fourier transform.

A continuous signal $x(t)$ can be represented as a sum of complex sinusoids with different frequencies.

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

- $x(t)$: signal in the time domain
- $X(\omega)$: frequency representation
- ω : continuous angular frequency

The inverse transform is

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

So,

$$x(t) \longleftrightarrow X(\omega)$$

Vector space		Fourier analysis
$\mathbf{x}$	$\leftrightarrow$	$x(t)$
$\mathbf{b}_k$	$\leftrightarrow$	$e^{j\omega t}$
a_k	$\leftrightarrow$	$X(\omega)$
$\langle \mathbf{x}, \mathbf{b}_k \rangle$	$\leftrightarrow$	$\int x(t) e^{-j\omega t} dt$
$\sum_k a_k \mathbf{b}_k$	$\leftrightarrow$	$\frac{1}{2\pi} \int X(\omega) e^{j\omega t} d\omega$

From Continuous-time Fourier Transform to DTFT

- Continuous-time Fourier Transform

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

Ω : continuous-time angular frequency, unit: rad/s
 T_s : sample interval, unit:
: integer sample index

- Sampling the signal in time

- Suppose the signals is sampled every T_s seconds

$$t = nT_s$$

- Then, $x[n] = x(nT_s)$
- The continuous signal $x(t)$ is now represented by discrete sample $x[n]$

$$\cdots, x[-1], x[0], x[1], x[2], \cdots$$

From Continuous-time Fourier Transform to DTFT

- The continuous integral can be viewed as the limit of a weighted summation

$$X_c(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$



Approximate the integral by a Riemann sum with sampling interval T_s

$$X_c(\Omega) \approx T_s \sum_{n=-\infty}^{\infty} x(nT_s) e^{-j\Omega nT_s} \quad x[n] = x(nT_s), w = \Omega T_s$$

- Define the discrete-time frequency
 - Let $w = \Omega T_s$, where w is the **normalized angular frequency** in **rad/sample**
 - Then

$$e^{-j\Omega nT_s} = e^{-jwn} \text{ and } x(nT_s) = x[n]$$

$$X_d(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$$

$$X_c(\Omega) \approx T_s X_d(e^{-jwn}), w = \Omega T_s$$

$$\Omega = 2\pi f \text{ (rad/s)}$$

$$\text{After sampling } w = \Omega T_s, \text{ where } T_s = \frac{1}{f_s}$$

$$\text{Therefore, } w = \Omega \frac{1}{f_s} = 2\pi \frac{f}{f_s}$$

1-D Discrete-time Fourier Transform

- Given a 1-d sequence $s[k], k \in \mathbf{Z} = \{\dots, -1, 0, 1, 2, 3, \dots\}$

- Fourier transform

$$S(e^{j\omega}) = \sum_{k=-\infty}^{\infty} s[k]e^{-j\omega k} \quad \omega \in \mathbb{R}$$

- Fourier transform is periodic with 2π , $S(e^{j(\omega+2\pi)}) = S(e^{j\omega})$

- $S(e^{j(\omega+2\pi)}) = \sum_{k=-\infty}^{\infty} s[k]e^{-j\omega k} = \sum_{k=-\infty}^{\infty} s[k]e^{-j(\omega+2\pi)k} = \sum_{k=-\infty}^{\infty} s[k]e^{-j\omega k}e^{-j2\pi k}$

- Inverse Fourier transform

$$s[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S(e^{j\omega})e^{j\omega k} d\omega \quad \omega \in \mathbb{R}$$

2-D Discrete-time Fourier Transform

Given a 2-d array of image samples

$$s[m, n], \quad m, n \in \mathbf{Z}^2$$

Fourier transform

$$S(e^{j\omega_x}, e^{j\omega_y}) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[m, n] e^{-j\omega_x m - j\omega_y n} \quad \omega_x, \omega_y \in \mathbb{R}^2$$

Fourier transform is 2π -periodic both in ω_x and ω_y

Inverse Fourier transform

$$s[m, n] = \frac{1}{4\pi^2} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} S(e^{j\omega_x}, e^{j\omega_y}) e^{j\omega_x m + j\omega_y n} d\omega_x d\omega_y$$

2-D Discrete-time Fourier Transform

- Define normalized frequency variables
 - Let $u = w_x/2\pi$, $v = w_y/2\pi$, where $w_x, w_y \in \mathbb{R}$
- Substitute into the DTFT

$$S(e^{j2\pi u}, e^{j2\pi v}) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[m, n] e^{-j(2\pi um + 2\pi vn)} \quad S(u, v) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[m, n] e^{-j(um + vn)}$$

$$\begin{array}{ll} w_x \text{ [rad/sample]} & \leftrightarrow \mu \text{ [cycles/sample]} \\ w_y \text{ [rad/sample]} & \leftrightarrow v \text{ [cycles/sample]} \end{array} \quad \text{with } u, v \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

2-D Discrete-time Fourier Transform

- Inverse Fourier transform

$$s[m, n] = \frac{1}{4\pi^2} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} S(e^{j\omega_x}, e^{j\omega_y}) e^{j\omega_x m + j\omega_y n} d\omega_x d\omega_y$$

Using $\omega_x = 2\pi u$, $\omega_y = 2\pi v$ and $d\omega_x d\omega_y = (2\pi)^2 du dv$

$$s[m, n] = \int_{-1/2}^{1/2} \int_{-1/2}^{1/2} S(u, v) e^{j2\pi(um+vn)} du dv$$

Fourier Transform

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

M, N : height and width of the image

Fourier Spectrum

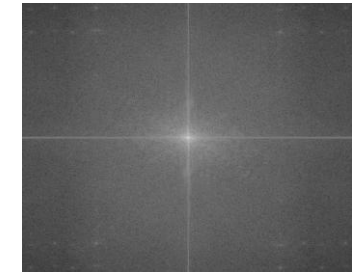
$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} = A(u, v) + jB(u, v)$$

Magnitude: $|F(u, v)| = \sqrt{A(u, v)^2 + B(u, v)^2}$

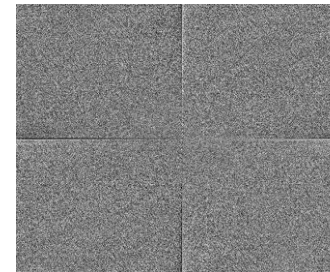
Phase: $\phi(u, v) = \angle F(u, v) = \text{atan2}(B(u, v), A(u, v))$



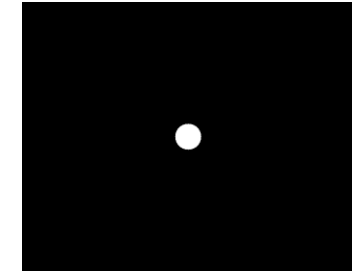
Original Image



Magnitude Spectrum

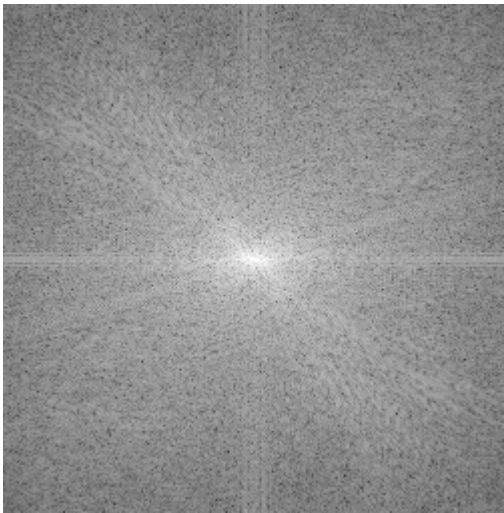


Phase Spectrum



Low-frequency Region

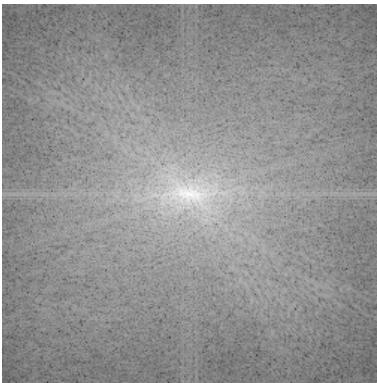
Fourier Spectrum



Magnitude spectrum



FT



Inverse
FT

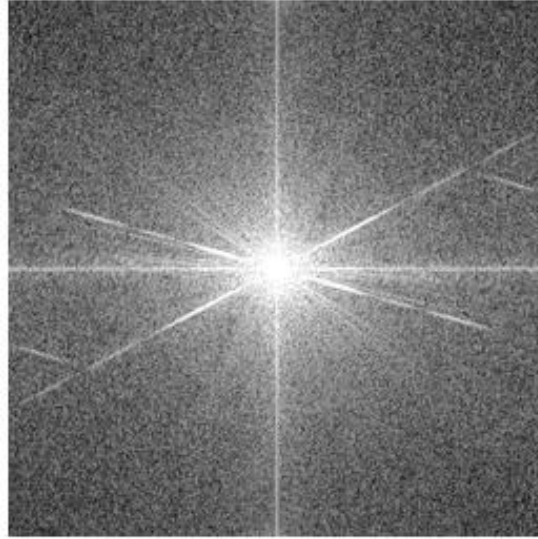


Fourier Spectrum

Original image



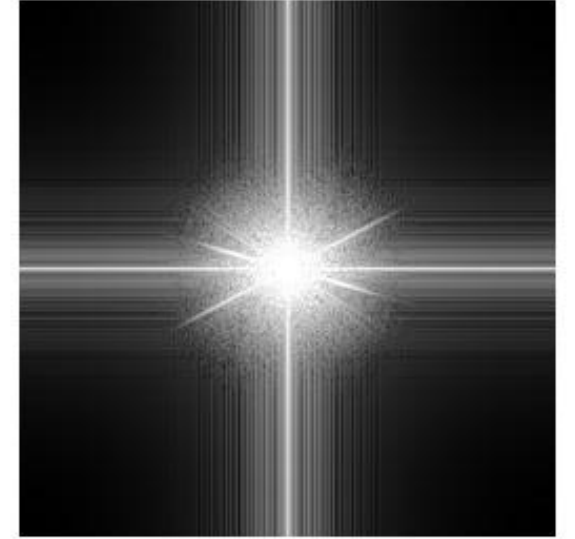
Original spectrum



Gaussian blur image



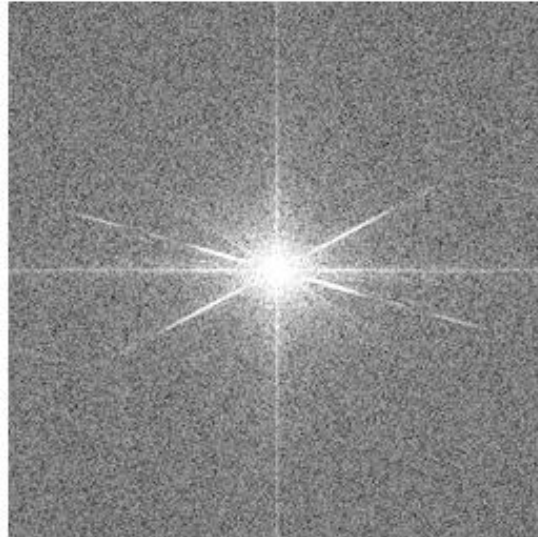
Gaussian blur spectrum



Gaussian noise image



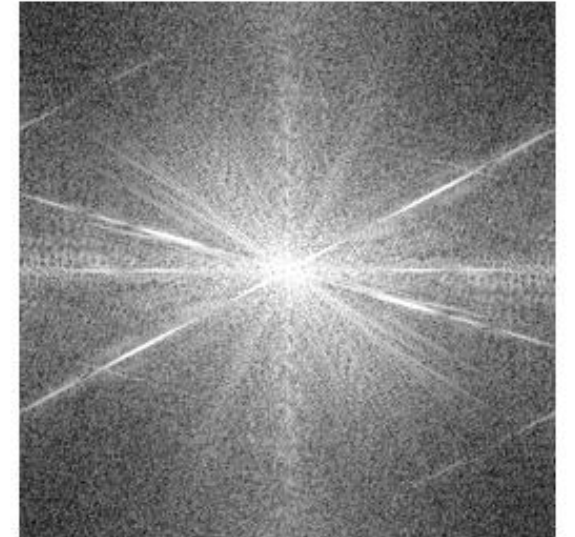
Gaussian noise spectrum



Sobel edge image



Sobel edge spectrum



FT Property: Convolution in the Fourier Domain

- Spatial domain

$$g[m, n] = f[m, n] * h[m, n]$$

- Fourier domain

$$G[u, v] = F[u, v]H[u, v]$$

- Spatial domain

$$g[m, n] = f[m, n]h[m, n]$$

- Fourier domain

$$G[u, v] = F[u, v] * H[u, v]$$

Convolution in
Spatial Domain



Multiplication in
Frequency Domain

FT Property

- Differentiation

- $F\left[\frac{d(x[n]-x[n-1])}{dt}\right] = (1 - e^{-jw})X(e^{jw}) = 2\left|\sin\frac{w}{2}\right|X(e^{jw})$

Differentiation emphasizes high-frequency components because its Fourier transform multiplies the spectrum by w .

- Spatial (Time) shift

- $x[n - n_0] \Leftrightarrow X(e^{jw})e^{-jwn_0}$

Fourier magnitude provides a shift-invariant representation.

- Convolution

- $x[n] * h[n] \Leftrightarrow X(e^{jw})H(e^{jw})$

Convolution becomes multiplication, making frequency-domain filtering simple and efficient.

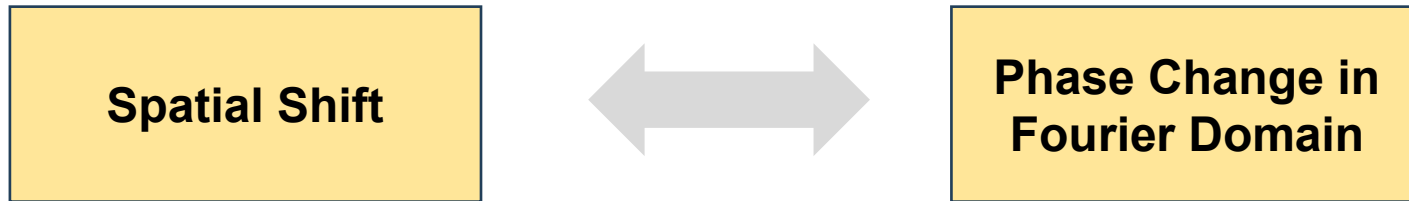
FT Property: Spatial Shift Affects Phase, not Magnitude

- If an image is shifted by (m_0, n_0)

$$f[m - m_0, n - n_0]$$

- Its Fourier transform becomes

$$F(u, v)e^{-j(um_0 + vn_0)}$$



Proof of the Convolution Theorem

Let

$$g[m, n] = f[m, n] * h[m, n]$$

where

$$g[m, n] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] h[m - k, n - l].$$

Take the 2-D Fourier transform:

$$G(u, v) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} g[m, n] e^{-j(um+vn)}.$$

Substituting the convolution expression,

$$G(u, v) = \sum_m \sum_n \left[\sum_k \sum_l f[k, l] h[m - k, n - l] \right] e^{-j(um+vn)}.$$

Rearranging the summations,

$$G(u, v) = \sum_k \sum_l f[k, l] \sum_m \sum_n h[m - k, n - l] e^{-j(um+vn)}.$$

Now define

$$p = m - k, \quad q = n - l$$

so that

$$m = p + k, \quad n = q + l.$$

Then,

$$G(u, v) = \sum_k \sum_l f[k, l] \sum_p \sum_q h[p, q] e^{-j[u(p+k)+v(q+l)]}.$$

Separate the exponential term:

$$e^{-j[u(p+k)+v(q+l)]} = e^{-j(uk+vl)} e^{-j(up+vq)}.$$

Therefore,

$$G(u, v) = \left[\sum_k \sum_l f[k, l] e^{-j(uk+vl)} \right] \left[\sum_p \sum_q h[p, q] e^{-j(up+vq)} \right]$$

Recognizing the two Fourier transforms,

$$\boxed{G(u, v) = F(u, v) H(u, v)}$$

Hence,

$$\boxed{f[m, n] * h[m, n] \iff F(u, v) H(u, v)}$$

Proof of the Differentiation Property (Continuous-Time)

Let

$$y(t) = \frac{dx(t)}{dt}.$$

Take the Fourier transform:

$$Y(j\Omega) = \int_{-\infty}^{\infty} \frac{dx(t)}{dt} e^{-j\Omega t} dt.$$

Using integration by parts, let

$$u = e^{-j\Omega t}, \quad dv = \frac{dx(t)}{dt} dt.$$

Then,

$$du = -j\Omega e^{-j\Omega t} dt, \quad v = x(t).$$

So,

$$Y(j\Omega) = \left[x(t) e^{-j\Omega t} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x(t) (-j\Omega e^{-j\Omega t}) dt.$$

Hence,

$$Y(j\Omega) = \left[x(t) e^{-j\Omega t} \right]_{-\infty}^{\infty} + j\Omega \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt.$$

Assuming

$$x(t) \rightarrow 0 \quad \text{as} \quad t \rightarrow \pm\infty,$$

the boundary term vanishes:

$$\left[x(t) e^{-j\Omega t} \right]_{-\infty}^{\infty} = 0.$$

Therefore,

$$Y(j\Omega) = j\Omega X(j\Omega).$$

Recognizing the Fourier transform of $x(t)$,

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt,$$

we obtain

$$\mathcal{F} \left\{ \frac{dx(t)}{dt} \right\} = j\Omega X(j\Omega)$$

and thus

$$|Y(j\Omega)| = |\Omega| |X(j\Omega)|.$$

Hence,

$$\frac{dx(t)}{dt} \iff j\Omega X(j\Omega)$$

Proof of the First-Difference Property (Discrete-Time)

Let

$$y[n] = x[n] - x[n - 1].$$

Take the DTFT:

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y[n]e^{-j\omega n}.$$

Substituting the difference expression,

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n] - x[n - 1])e^{-j\omega n}.$$

Rearranging the summation,

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} - \sum_{n=-\infty}^{\infty} x[n - 1]e^{-j\omega n}.$$

The first term is simply

$$\sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = X(e^{j\omega}).$$

For the second term, define

$$m = n - 1, \quad \text{so that} \quad n = m + 1.$$

Then,

$$\sum_{n=-\infty}^{\infty} x[n - 1]e^{-j\omega n} = \sum_{m=-\infty}^{\infty} x[m]e^{-j\omega(m+1)}.$$

Separate the exponential term:

$$e^{-j\omega(m+1)} = e^{-j\omega m}e^{-j\omega}.$$

Thus,

$$\sum_{n=-\infty}^{\infty} x[n - 1]e^{-j\omega n} = e^{-j\omega} \sum_{m=-\infty}^{\infty} x[m]e^{-j\omega m}.$$

Recognizing the DTFT of $x[n]$,

$$\sum_{m=-\infty}^{\infty} x[m]e^{-j\omega m} = X(e^{j\omega}),$$

we obtain

$$Y(e^{j\omega}) = X(e^{j\omega}) - e^{-j\omega}X(e^{j\omega}).$$

Therefore,

$$Y(e^{j\omega}) = (1 - e^{-j\omega})X(e^{j\omega}).$$

Hence,

$$\boxed{\mathcal{F}\{x[n] - x[n - 1]\} = (1 - e^{-j\omega})X(e^{j\omega})}$$

or equivalently,

$$\boxed{x[n] - x[n - 1] \iff (1 - e^{-j\omega})X(e^{j\omega})}$$

Magnitude Form of the Discrete-Time Difference

Now factor the frequency response:

$$1 - e^{-j\omega} = e^{-j\omega/2} \left(e^{j\omega/2} - e^{-j\omega/2} \right).$$

Using

$$e^{j\theta} - e^{-j\theta} = 2j \sin \theta,$$

we get

$$1 - e^{-j\omega} = 2je^{-j\omega/2} \sin \frac{\omega}{2}.$$

Therefore,

$$Y(e^{j\omega}) = 2je^{-j\omega/2} \sin \frac{\omega}{2} X(e^{j\omega}).$$

Taking magnitude on both sides,

$$|Y(e^{j\omega})| = \left| 2je^{-j\omega/2} \sin \frac{\omega}{2} \right| |X(e^{j\omega})|.$$

Since

$$|j| = 1, \quad |e^{-j\omega/2}| = 1,$$

we obtain

$$|Y(e^{j\omega})| = 2 \left| \sin \frac{\omega}{2} \right| |X(e^{j\omega})|$$

Hence,

$$|1 - e^{-j\omega}| = 2 \left| \sin \frac{\omega}{2} \right|$$

and finally

$$x[n] - x[n-1] \iff (1 - e^{-j\omega})X(e^{j\omega}), \quad |Y(e^{j\omega})| = 2 \left| \sin \frac{\omega}{2} \right| |X(e^{j\omega})|$$

If the Sampling Interval T_s is Included

If we define the discrete derivative approximation as

$$y[n] = \frac{x[n] - x[n-1]}{T_s},$$

then

$$Y(e^{j\omega}) = \frac{1 - e^{-j\omega}}{T_s} X(e^{j\omega})$$

and

$$|Y(e^{j\omega})| = \frac{2}{T_s} \left| \sin \frac{\omega}{2} \right| |X(e^{j\omega})|$$

5x5 Box-filter Revisited

$$f[m, n]$$



Original
Bike

$$g[m, n] = f * h$$



Bike blurred by convolution
Impulse response „box filter“

Since convolution in the spatial domain becomes multiplication in the Fourier domain, the box filter can be interpreted as a low-pass filter that attenuates high frequencies.

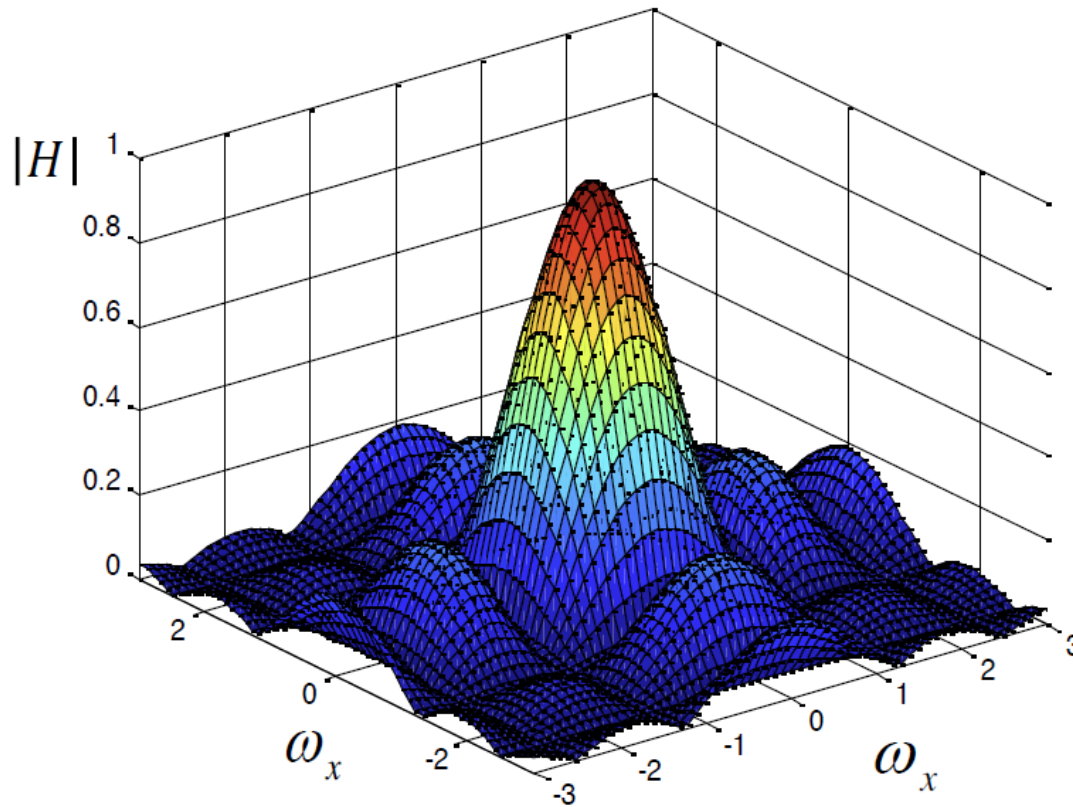
$$h[m, n] = \frac{1}{25} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned} & \frac{1}{25} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \left(\frac{1}{5} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right) \left(\frac{1}{5} [1 \quad 1 \quad 1 \quad 1 \quad 1] \right) \end{aligned}$$

$$g[m, n] = f[m, n] * h[m, n]$$

$$G(u, v) = F(u, v)H(u, v)$$

Frequency Response of 5x5 Lowpass Filter



$$\begin{aligned}
 H(e^{j\omega_x}, e^{j\omega_y}) &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} h[m, n] e^{-j\omega_x m - j\omega_y n} & h[m, n] &= \frac{1}{25}, \quad m, n \in \{-2, -1, 0, 1, 2\} \\
 &= \frac{1}{25} \sum_{m=-2}^2 \sum_{n=-2}^2 e^{-j\omega_x m - j\omega_y n} = \frac{1}{25} \sum_{m=-2}^2 e^{-j\omega_x m} \sum_{n=-2}^2 e^{-j\omega_y n} \\
 &= \frac{1}{25} \left(1 + 2\cos\omega_x + 2\cos(2\omega_x) \right) \left(1 + 2\cos\omega_y + 2\cos(2\omega_y) \right)
 \end{aligned}$$

Separable filter:
1-d frequency responses
are multiplied

The 5×5 box filter has a sinc-like frequency response: low frequencies are preserved, while many high-frequency components are attenuated

A separable 2-D filter can be decomposed into two 1-D filters; therefore, its 2-D frequency response is the product of the horizontal and vertical 1-D frequency responses.

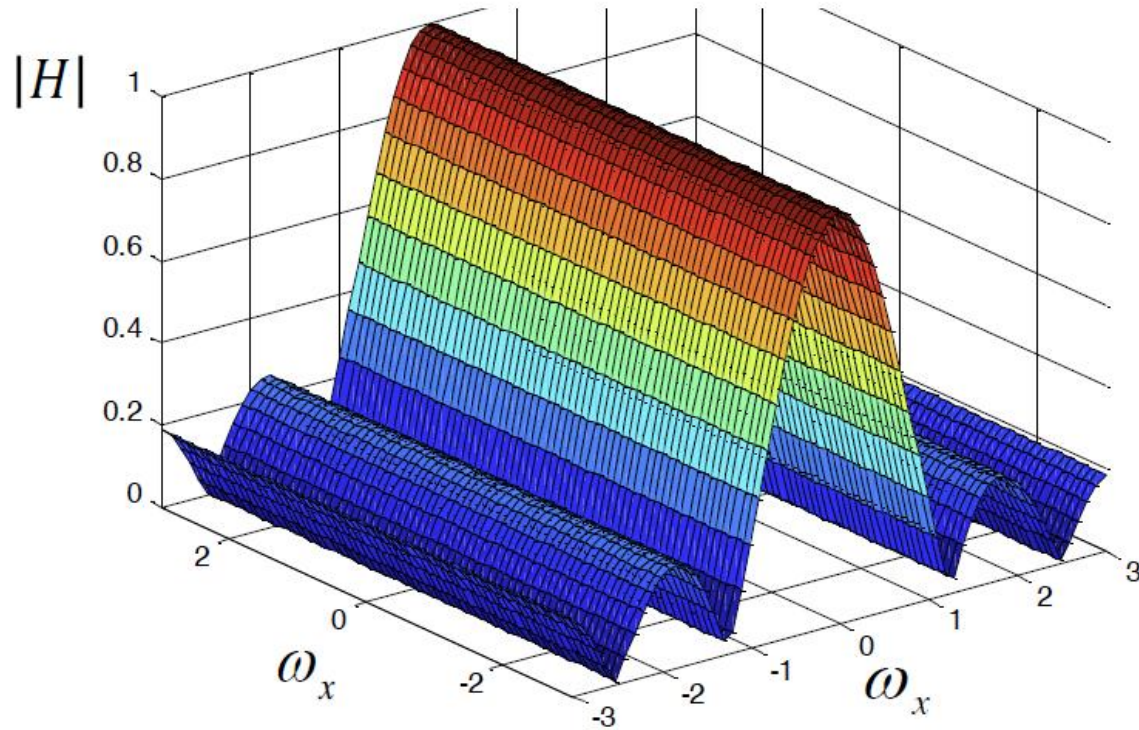
Horizontal Lowpass Filter



Bike blurred horizontally
Filter impulse response

$$\frac{1}{5} \begin{pmatrix} 1 & 1 & [1] & 1 & 1 \end{pmatrix}$$

$$H(e^{j\omega_x}, e^{j\omega_y}) = \frac{1}{5} (1 + 2\cos\omega_x + 2\cos(2\omega_x))$$



Sharpening Filter

$$f[m, n]$$



Original

$$g[m, n] = f * h$$



Sharpened

Filter impulse response

$$h[m, n] = \frac{1}{4} \begin{bmatrix} 0 & -1 & 0 \\ -1 & 8 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

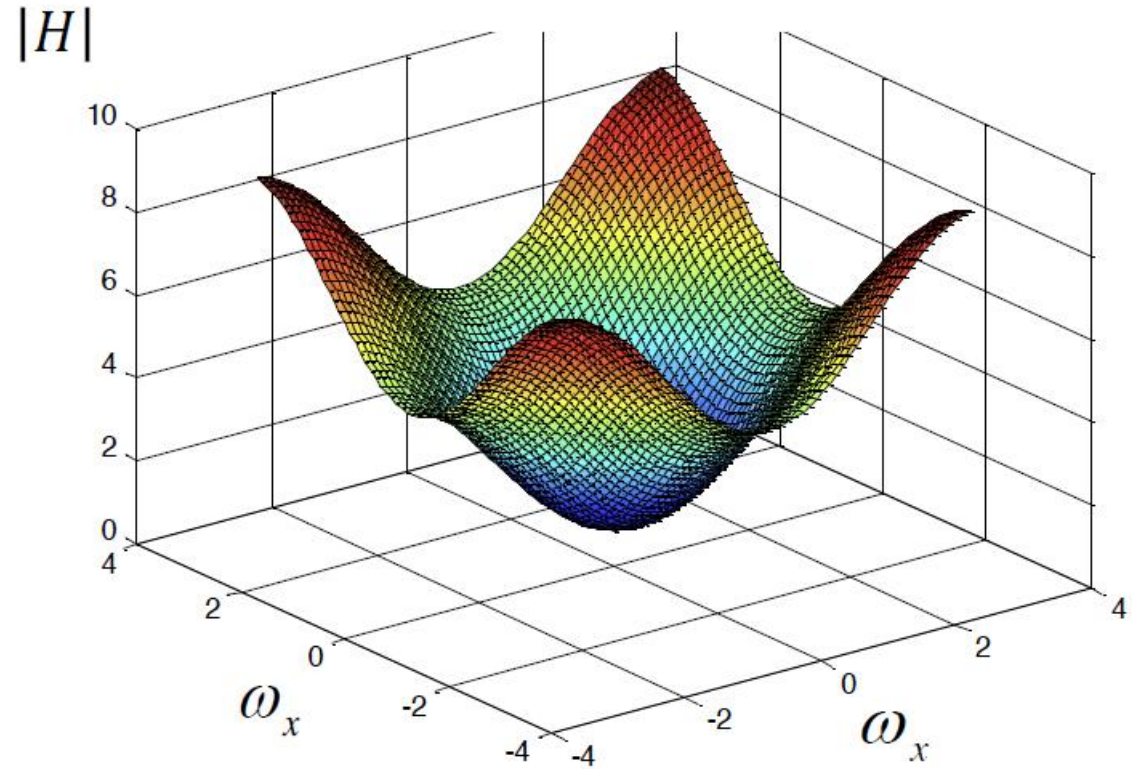
More Aggressive Sharpening



Bike sharpened
Filter impulse response

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & [5] & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

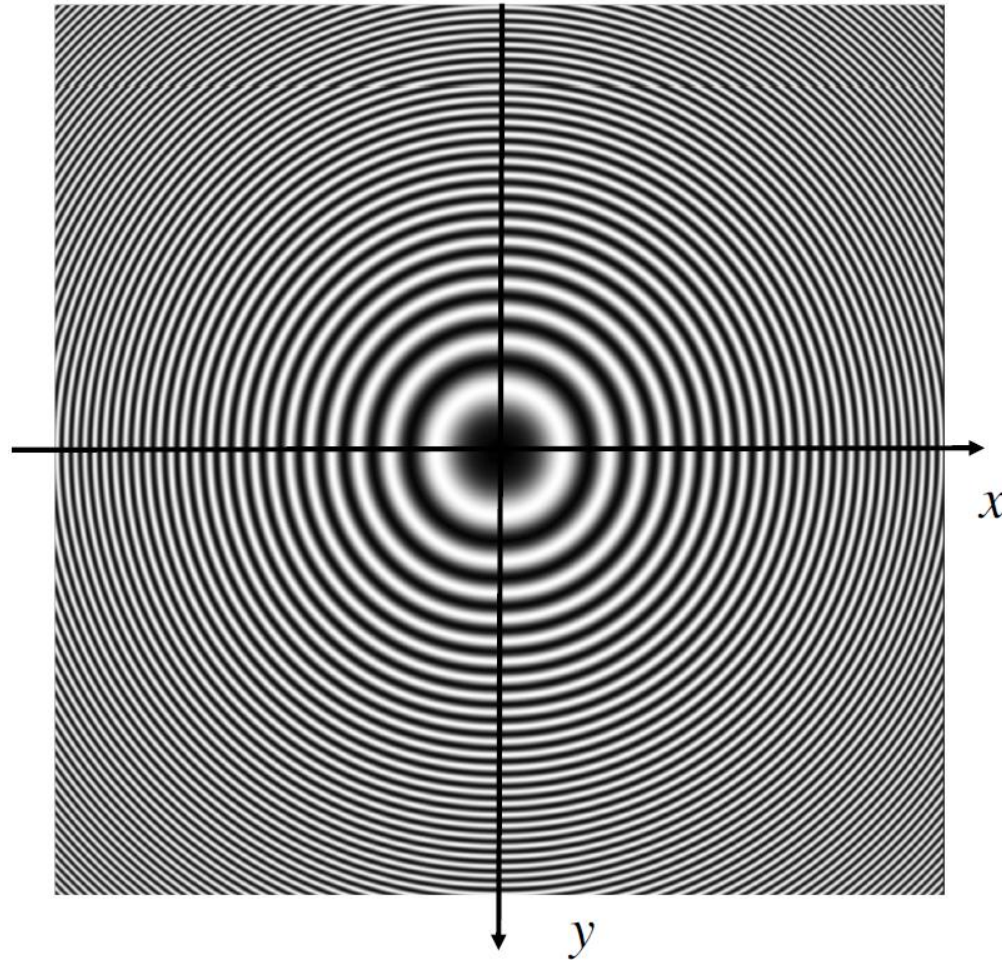
$$H(e^{j\omega_x}, e^{j\omega_y}) = 5 - 2\cos\omega_x - 2\cos\omega_y$$



Homework

- What about frequency response of Gaussian and Laplacian filter response?

Zoneplate Pattern to Visualize Frequency Plane



Equation to generate pattern

$$s(x, y) = \hat{s} \cos(a_x x^2 + a_y y^2) + s_0$$

Local frequency at (x, y)

$$w_x = \frac{\partial(a_x x^2 + a_y y^2)}{\partial x} = 2a_x x$$

$$w_y = \frac{\partial(a_x x^2 + a_y y^2)}{\partial y} = 2a_y y$$

$$s[x, y] \xrightleftharpoons[\text{Sampling}]{\text{Interpolation}} s(x, y)$$

Sampling Interpretation of 2-d Discrete-space Fourier Transform

How is the Fourier transform of $s[m,n]$ related to the Fourier transform of the continuous signal

$$s(x,y) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[m,n] \delta_2(x-m, y-n)$$

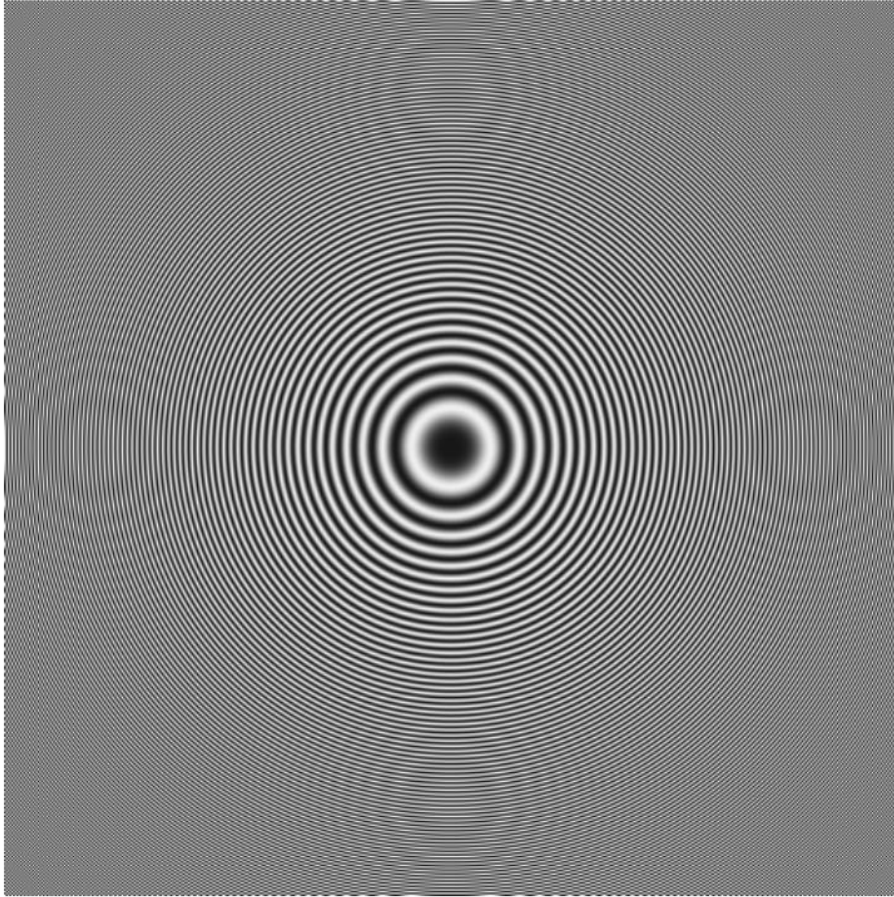
continuous discrete „continuous“
delta function

Continuous-space 2-d Fourier transform

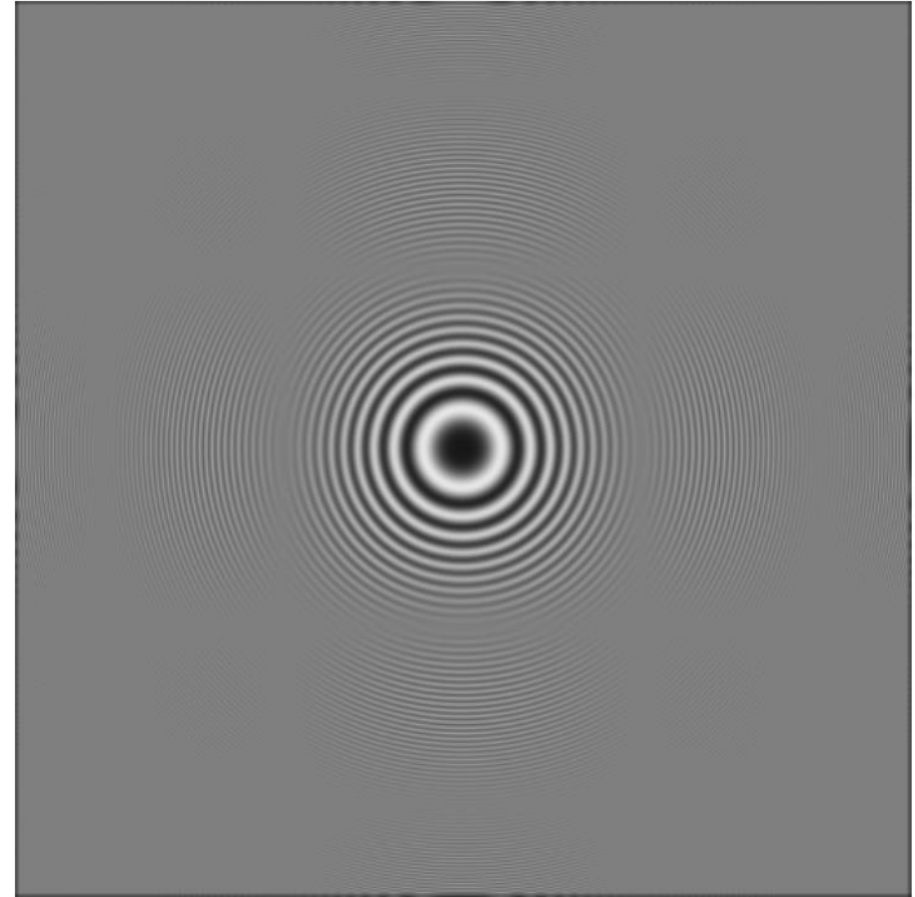
$$\begin{aligned} \hat{S}(\omega_x, \omega_y) &= \iint \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[m,n] \delta_2(x-m, y-n) e^{-j\omega_x x - j\omega_y y} dx dy \\ &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[m,n] e^{-j\omega_x m - j\omega_y n} = S(e^{j\omega_x}, e^{j\omega_y}) = S(e^{j(\omega_x + 2\pi)}, e^{j(\omega_y + 2\pi)}) \end{aligned}$$

- DTFT of $s[m,n]$ = continuous FT of its impulse-train representation
- $S(\omega_x, \omega_y)$ is 2π -periodic in both direction

Lowpass Filtering



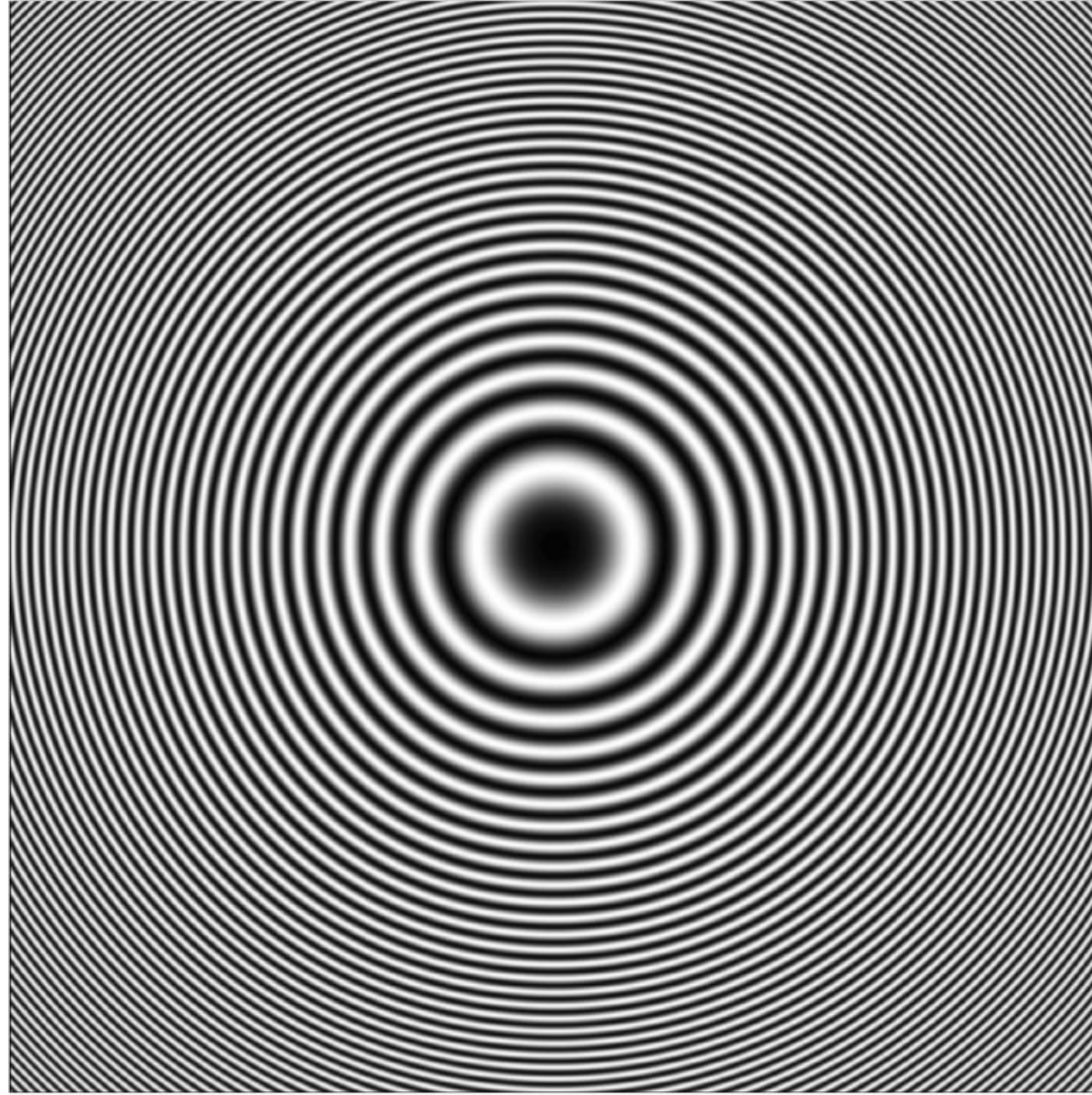
Original *Zoneplate* (512x512)



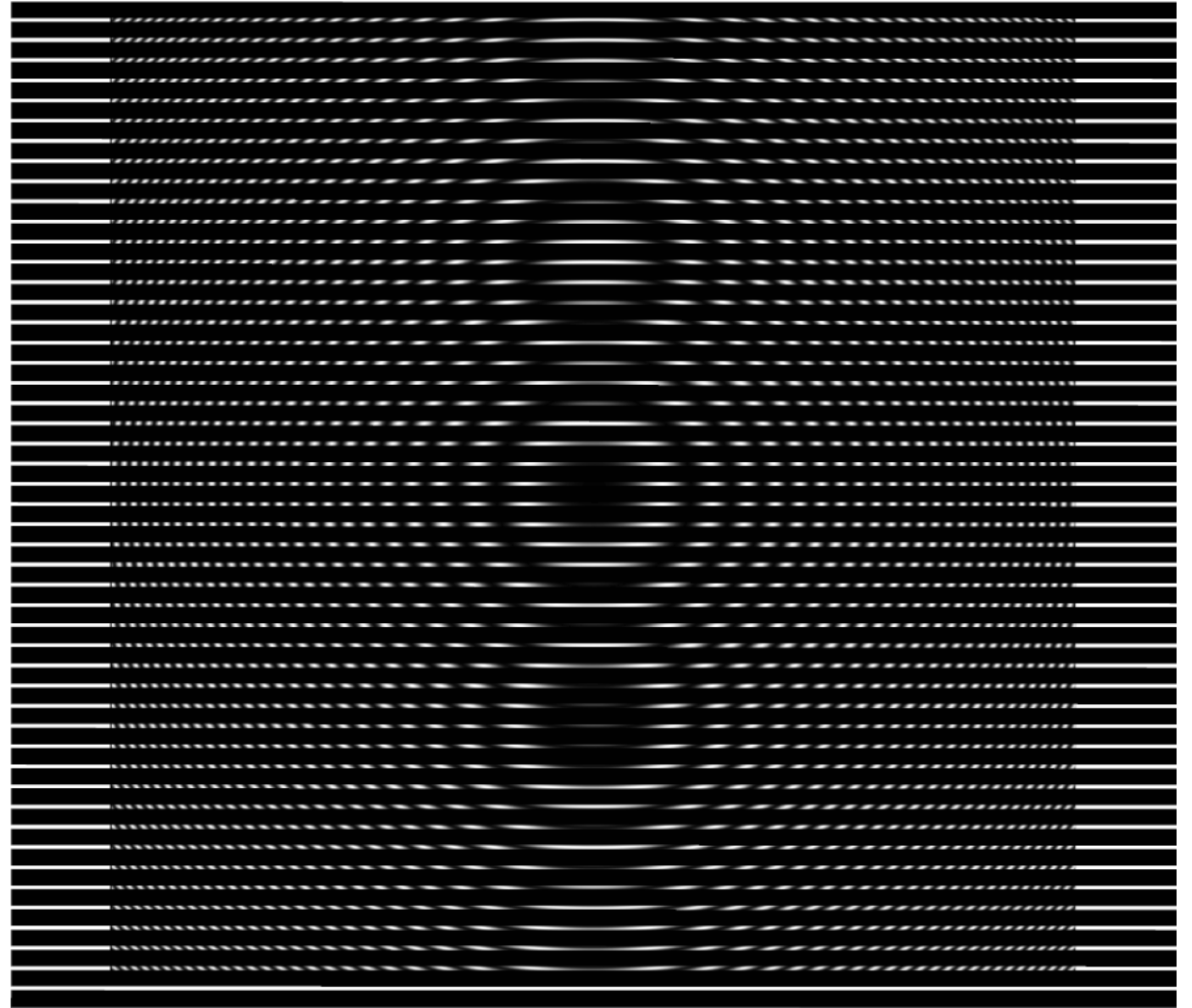
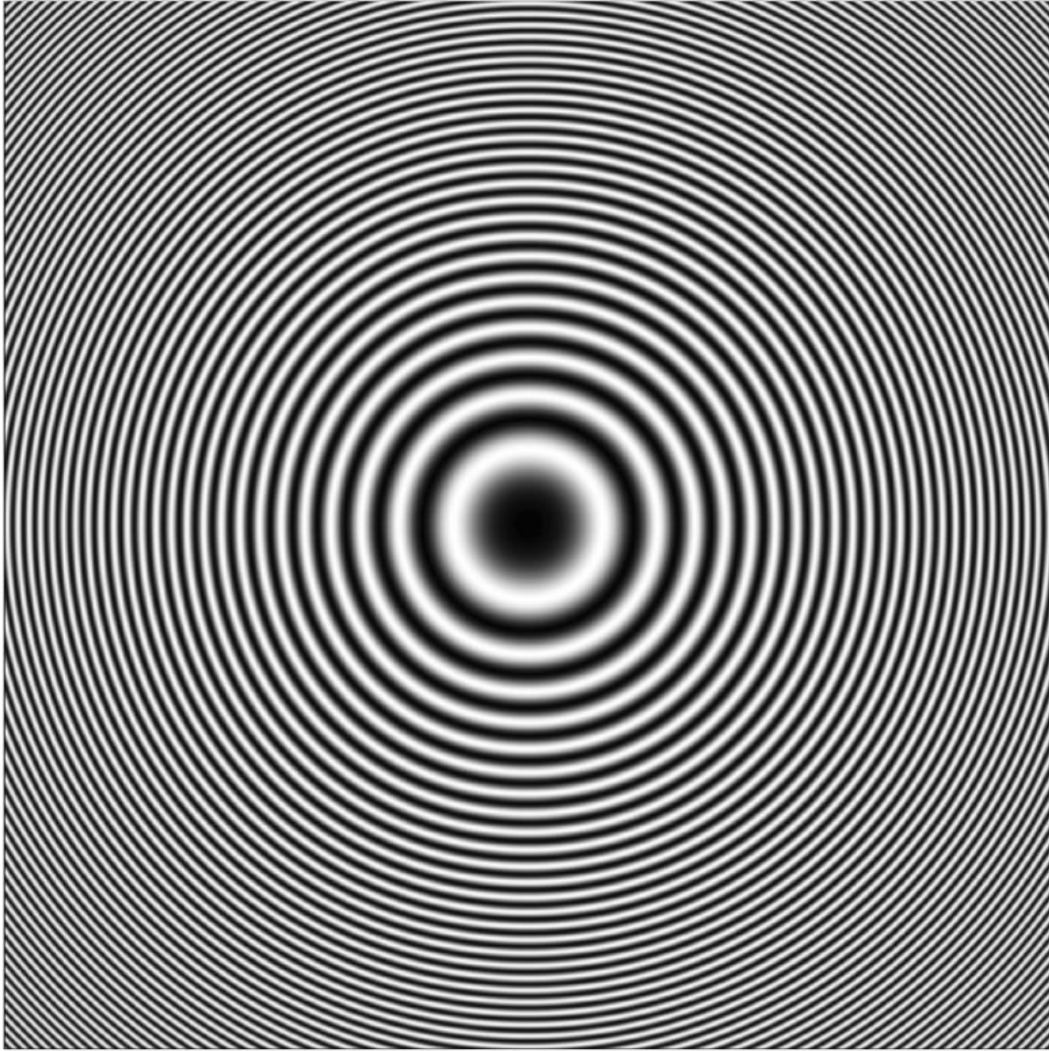
Lowpass filtered with 5x5 box filter

Low-frequency patterns near the center are preserved, while high-frequency patterns farther from the center are suppressed.

Zoneplate Demonstration: 1-D Sampling of 2-D signal



Zoneplate Demonstration: 1-D Sampling of 2-D signal



Zoneplate Demonstration: 1-D Sampling of 2-D signal

Zone Plate Phase

For a zone plate,

$$s(x, y) = \cos \phi(x, y)$$

with

$$\phi(x, y) = a_x x^2 + a_y y^2$$

The local angular frequency along the y -direction is

$$\omega_y(y) = \frac{\partial \phi}{\partial y} = 2a_y y$$

Therefore,

$$|y| \uparrow \Rightarrow |\omega_y| \uparrow$$

2:1 Downsampling Along the y -Direction

After downsampling by a factor of 2,

$$y = 2nT$$

and the sampled phase becomes

$$\phi_n = a_x x^2 + a_y (2nT)^2$$

$$\phi_n = a_x x^2 + 4a_y T^2 n^2$$

The discrete local frequency along n is approximately

$$\omega_d(n) = \frac{d\phi_n}{dn}$$

thus,

$$\omega_d(n) = 8a_y T^2 n$$

Hence,

$$|n| \uparrow \Rightarrow |\omega_d| \uparrow$$

The underlying frequency still keeps increasing after downsampling.

Discrete Frequency Is 2π -Periodic

For a discrete signal,

$$e^{j(\omega+2\pi)n} = e^{j\omega n}$$

Therefore,

$$\omega, \quad \omega + 2\pi, \quad \omega + 4\pi, \dots$$

are indistinguishable after sampling.

The observed frequency is therefore

$$\omega_{\text{alias}} = \omega_d \pmod{2\pi}$$

Frequency Folding

The true local frequency continues to increase:

$$0 \rightarrow \frac{\pi}{2} \rightarrow \pi \rightarrow \frac{3\pi}{2} \rightarrow 2\pi \rightarrow \frac{5\pi}{2} \rightarrow \dots$$

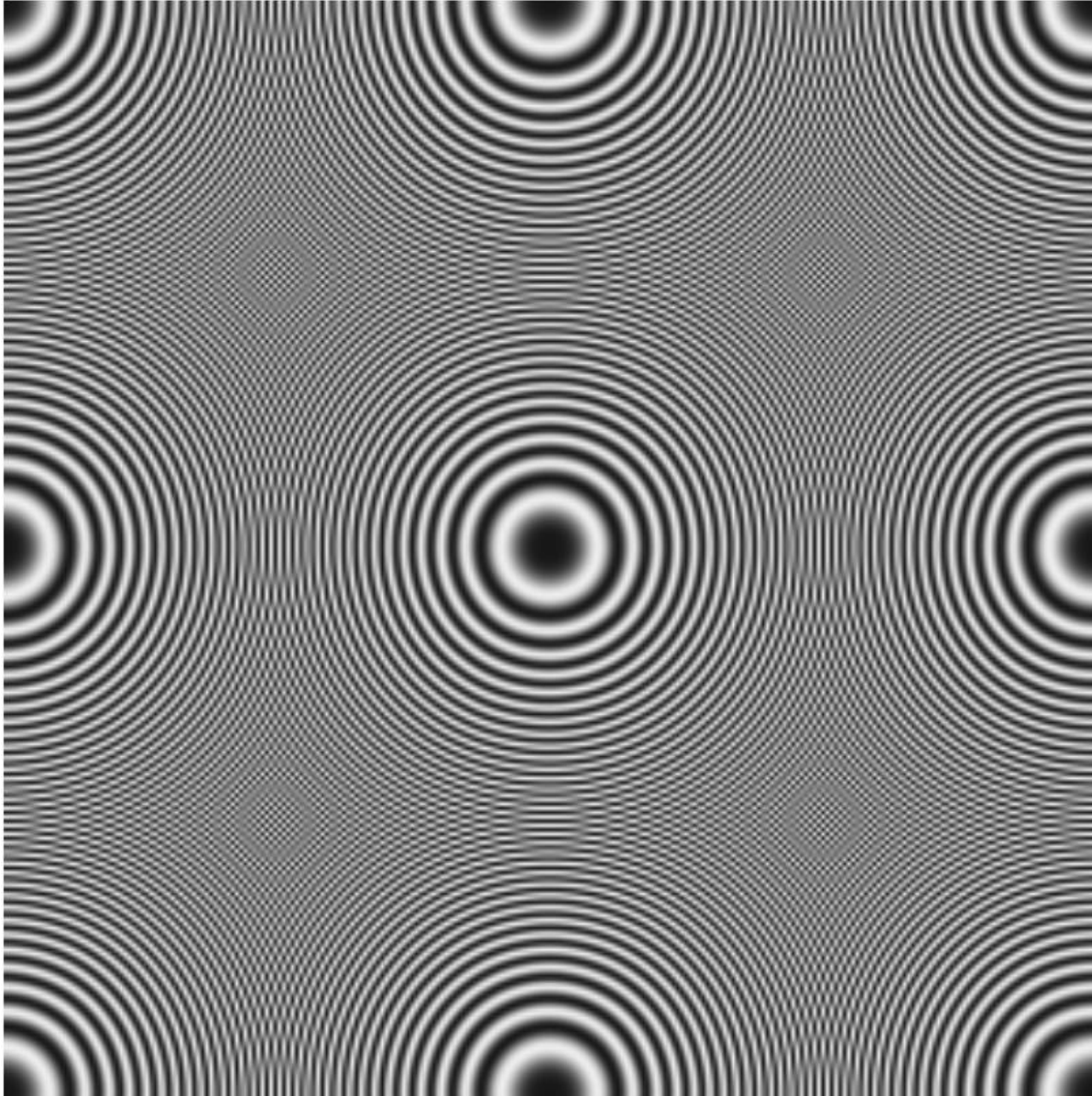
However, after sampling, it appears as

$$0 \rightarrow \frac{\pi}{2} \rightarrow \pi \rightarrow -\frac{\pi}{2} \rightarrow 0 \rightarrow \frac{\pi}{2} \rightarrow \dots$$

In terms of apparent frequency magnitude,

$$0 \rightarrow \pi \rightarrow 0 \rightarrow \pi \rightarrow 0 \rightarrow \dots$$

Horizontal/Vertical 2:1 Subsampling without Prefiltering



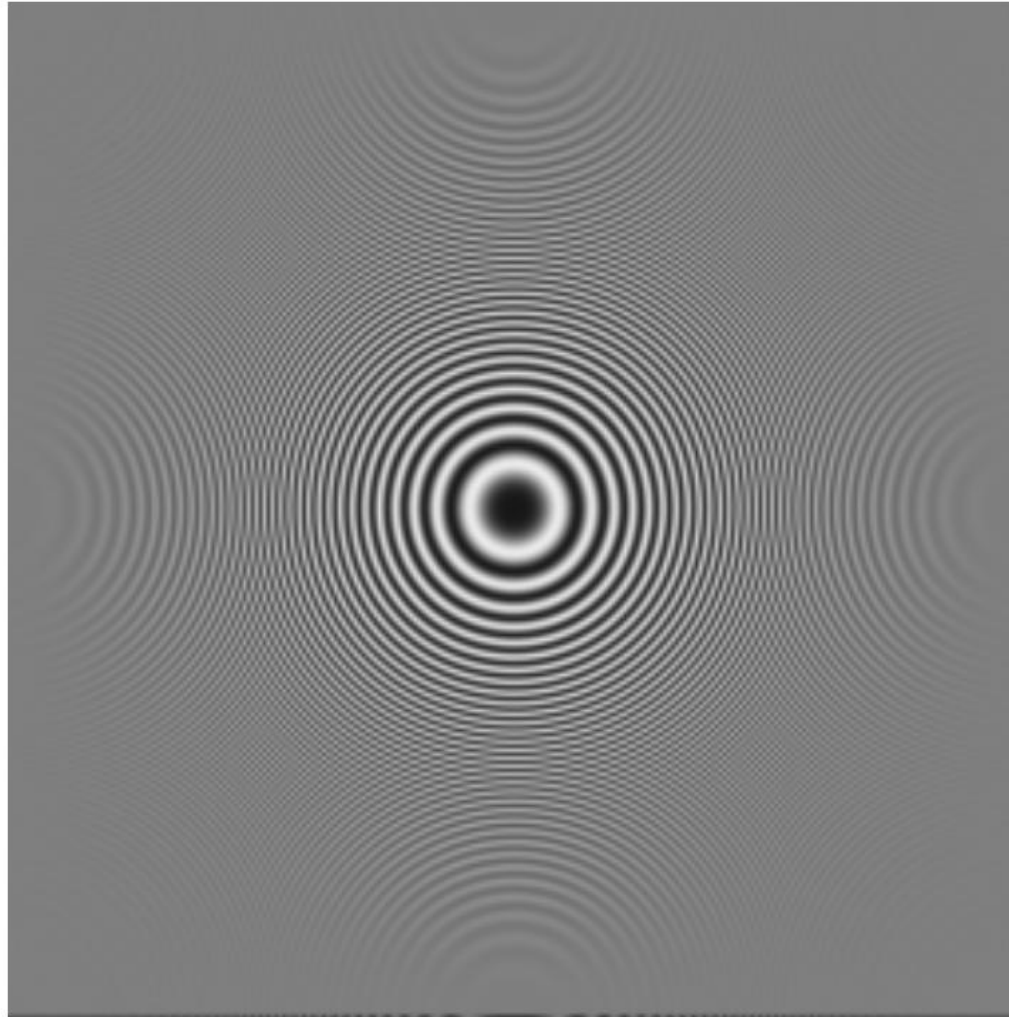
Without prefiltering, high-frequency components above the new Nyquist limit are folded into lower frequencies, producing aliasing.

Horizontal/Vertical 2:1 Subsampling with Prefiltering

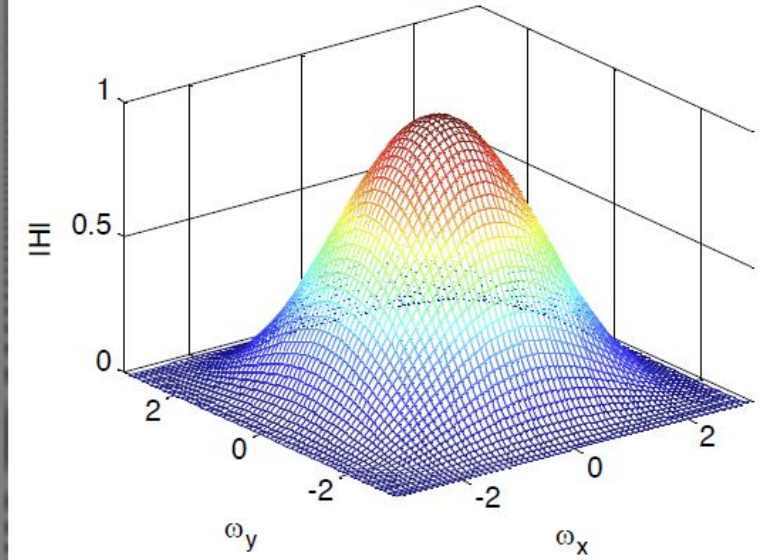
2d impulse response

$$\begin{pmatrix} \frac{1}{16} & \frac{1}{8} & \frac{1}{16} \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{8} \\ \frac{1}{16} & \frac{1}{8} & \frac{1}{16} \end{pmatrix}$$

$$\frac{1}{4} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \cdot \frac{1}{4} [1 \quad 2 \quad 1]$$



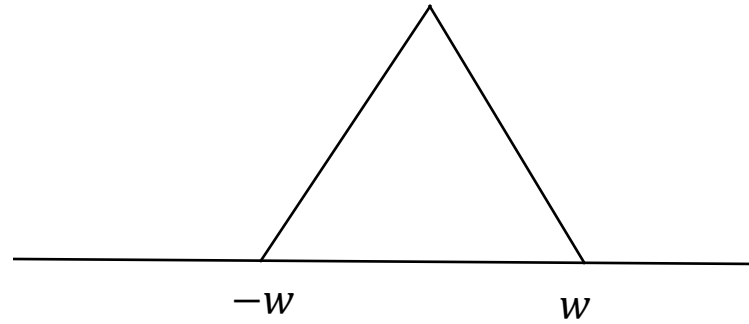
Frequency response



Digital Frequency After 2:1 Downsampling

- Digital frequency

- $w = 2\pi \frac{f}{f_s}$

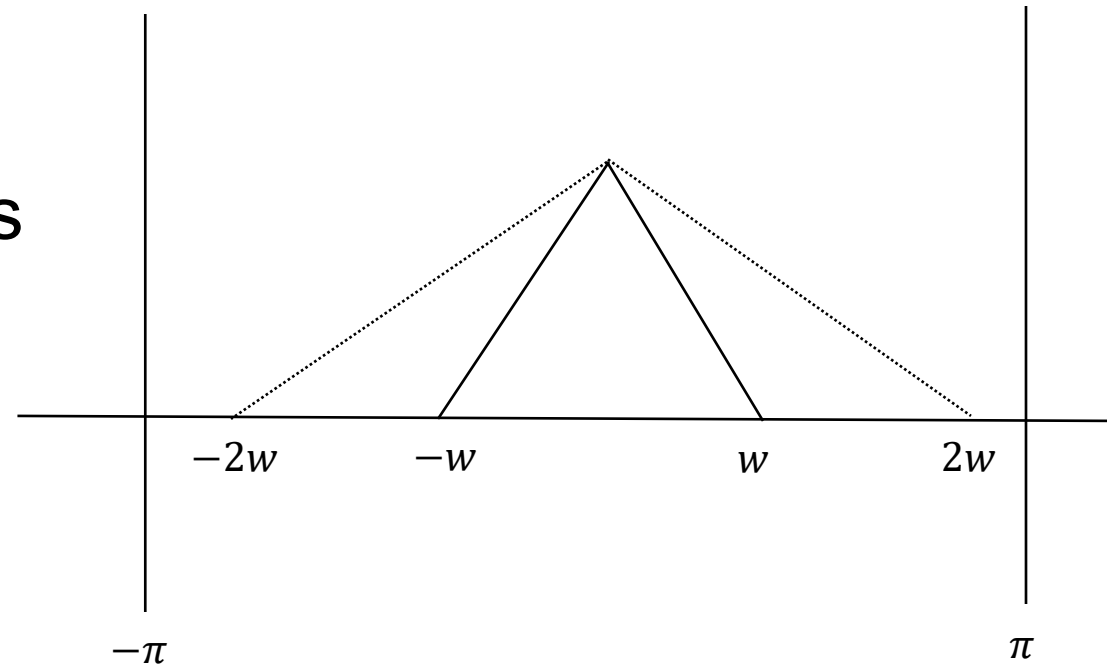


- 2:1 downsampling

- $f'_s = \frac{f_s}{2}$

- Then, digital frequency becomes

- $w' = 2\pi \frac{f}{f_s/2} = 2w$



Nyquist Limit after 2:1 Downsampling

For the original discrete image, the normalized angular frequency is defined as

$$w_x = \Omega_x T_x, w_y = \Omega_y T_y \quad (T_x \text{ and } T_y \text{ are spatial sampling interval}) \rightarrow \text{Original Nyquist limit is } |w_x|, |w_y| \leq \pi$$

After 2:1 Downsampling

Taking every other sample doubles the sampling interval:

$$T'_x = 2T_x, \quad T'_y = 2T_y$$

Thus, the sampling rate is reduced by half.

The physical Nyquist angular frequency becomes

$$\Omega'_N = \frac{\pi}{T'_x} = \frac{\pi}{2T_x}$$

and similarly for the y -direction.

Expressed in the Original Normalized-Frequency Scale

Using

$$\omega = \Omega T$$

with the original sampling interval T_x ,

$$\omega'_N = \Omega'_N T_x = \frac{\pi}{2T_x} T_x = \boxed{\frac{\pi}{2}}$$

Therefore,

$$\boxed{|w_x|, |w_y| \leq \frac{\pi}{2}}$$

Anti-Aliasing Condition for M:1 Downsampling

2:1 Downsampling

$$y[n] = x[2n]$$

An input frequency ω is mapped to

$$\omega \rightarrow 2\omega$$

after downsampling.

Nyquist Condition

To avoid aliasing,

$$|2\omega| \leq \pi$$

Therefore,

$$|\omega| \leq \frac{\pi}{2}$$

So, before 2:1 downsampling, the signal should be low-pass filtered to

$$-\frac{\pi}{2} \leq \omega \leq \frac{\pi}{2}$$

In Terms of Sampling Frequency

After downsampling,

$$f'_s = \frac{f_s}{2}$$

so the new Nyquist frequency is

$$f_{\max} = \frac{f'_s}{2} = \frac{f_s}{4}$$

General Case

For $M : 1$ downsampling,

$$|\omega| \leq \frac{\pi}{M}$$

Anti-aliasing low-pass filtering must be applied before downsampling.

Aliasing

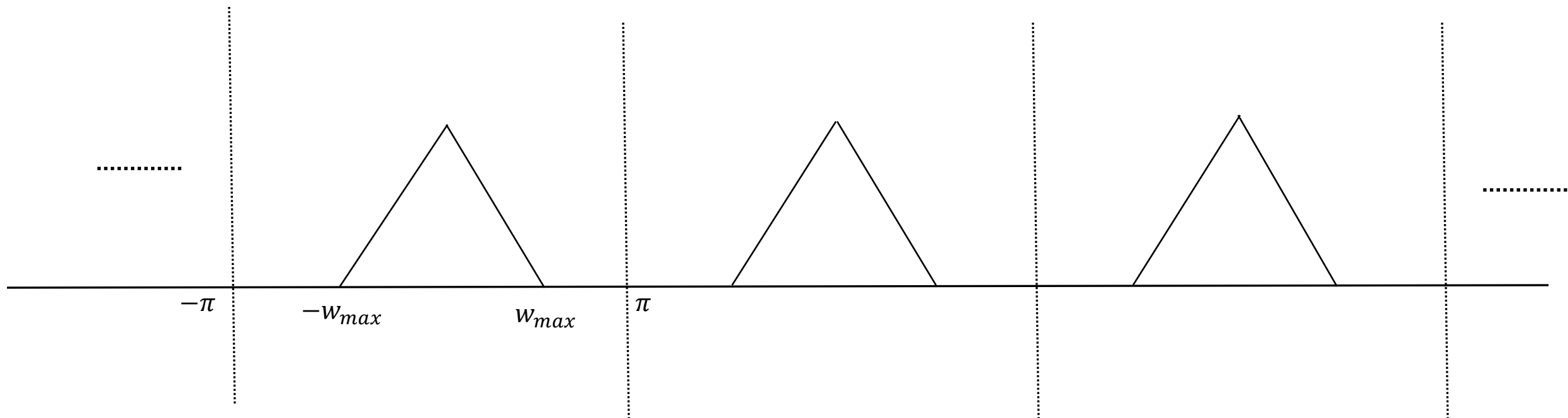
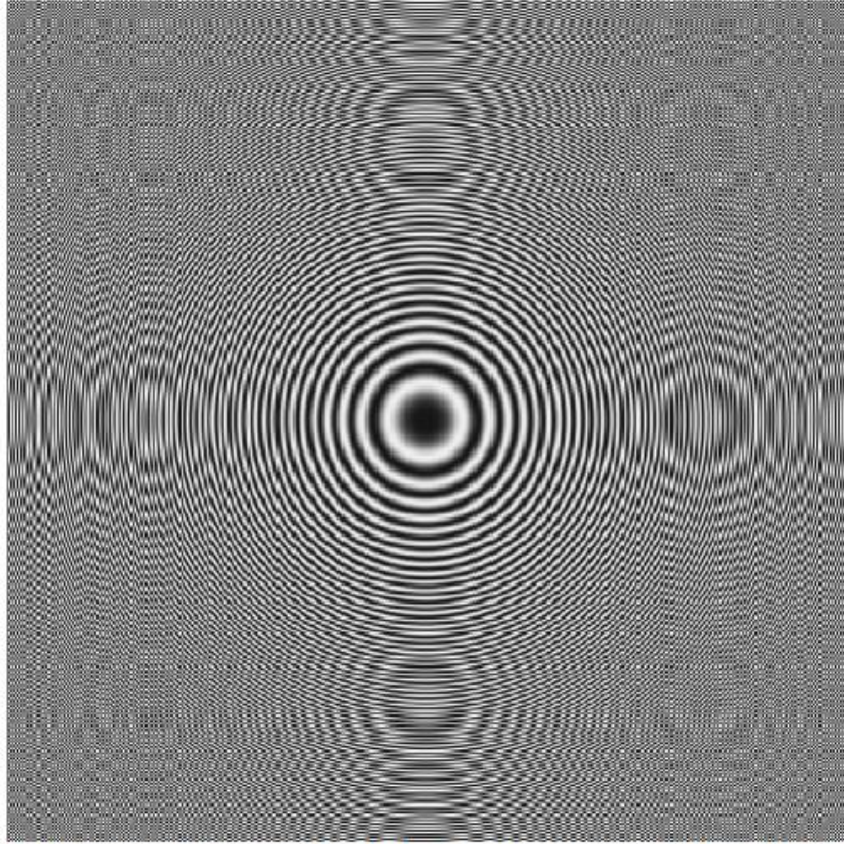
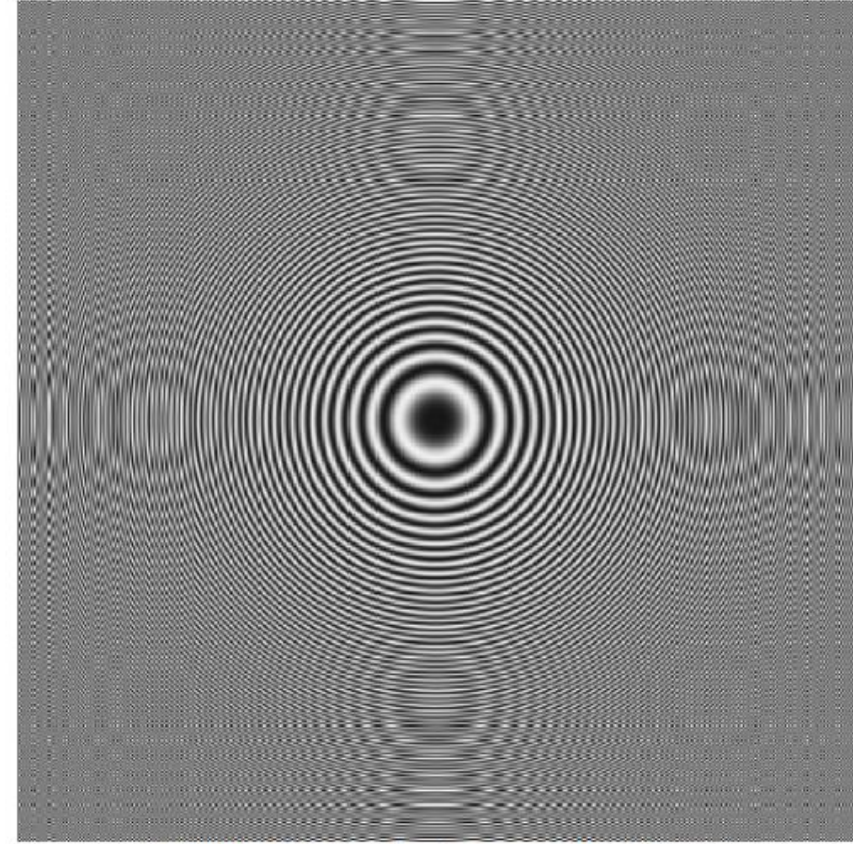


Image Magnification (10%)



Nearest neighbor interpolation



Bilinear interpolation

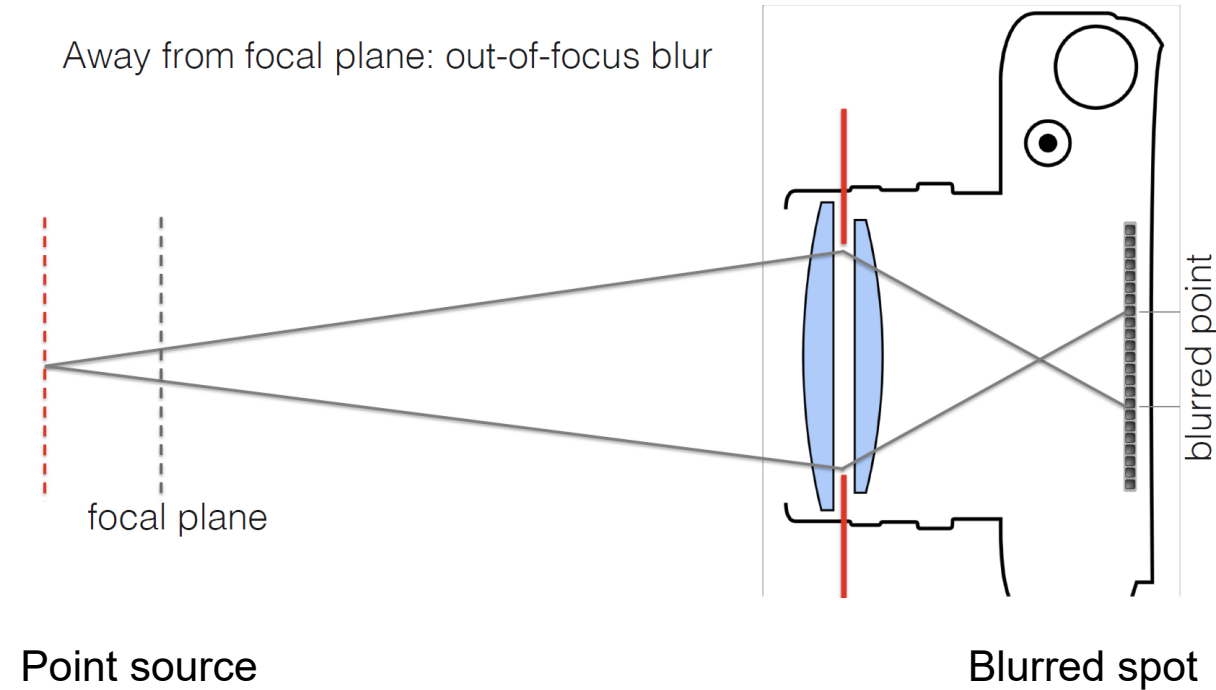
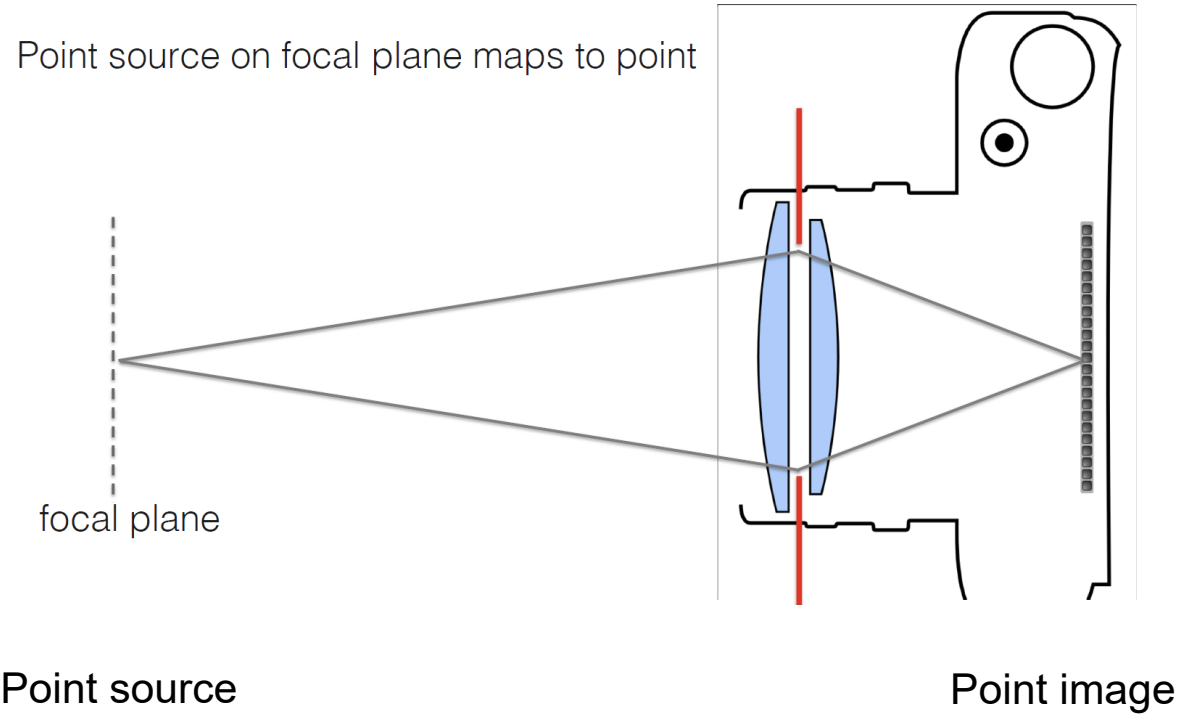
Blurry Images

$$g(x, y) = f(x, y) * h(x, y)$$

$f(x, y)$: Original clean image

$h(x, y)$: Point Spread Function (PSF)

PSF = A function that describes how a point source is spread in the imaging system.

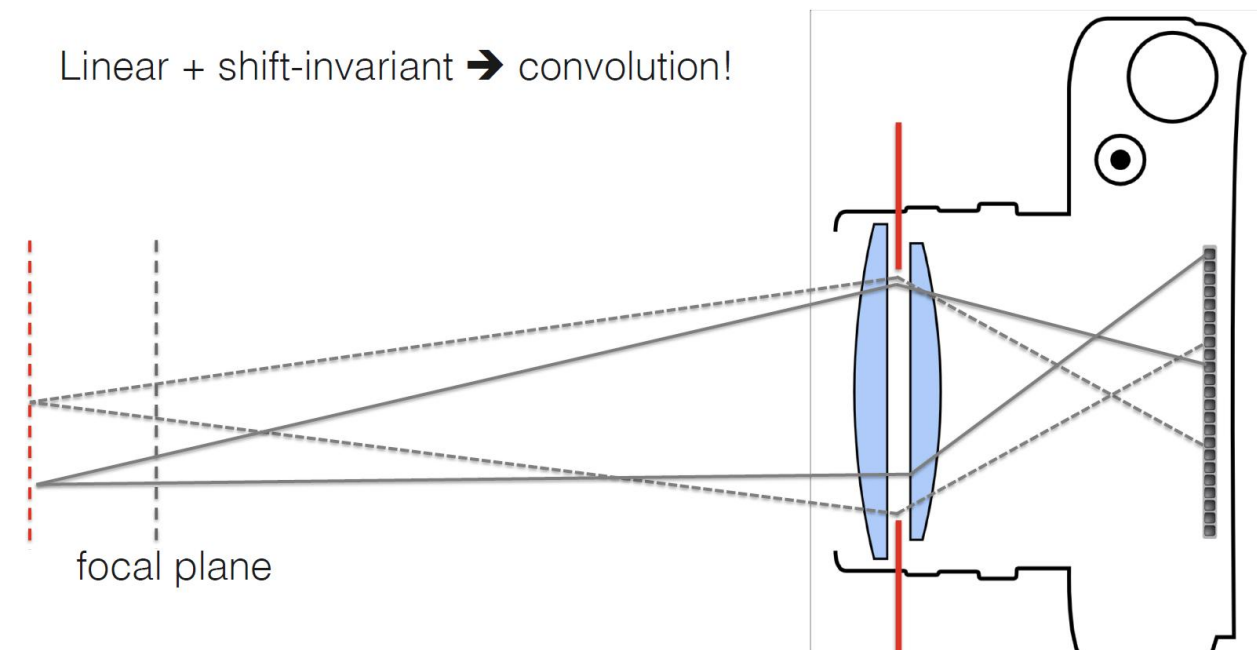


The focal plane is the object plane that is currently in focus. A point on this plane is mapped to a point on the sensor. A point away from this plane is mapped to a finite-sized blur spot.

Blurry Images

Linear + shift-invariant $\rightarrow$ convolution!

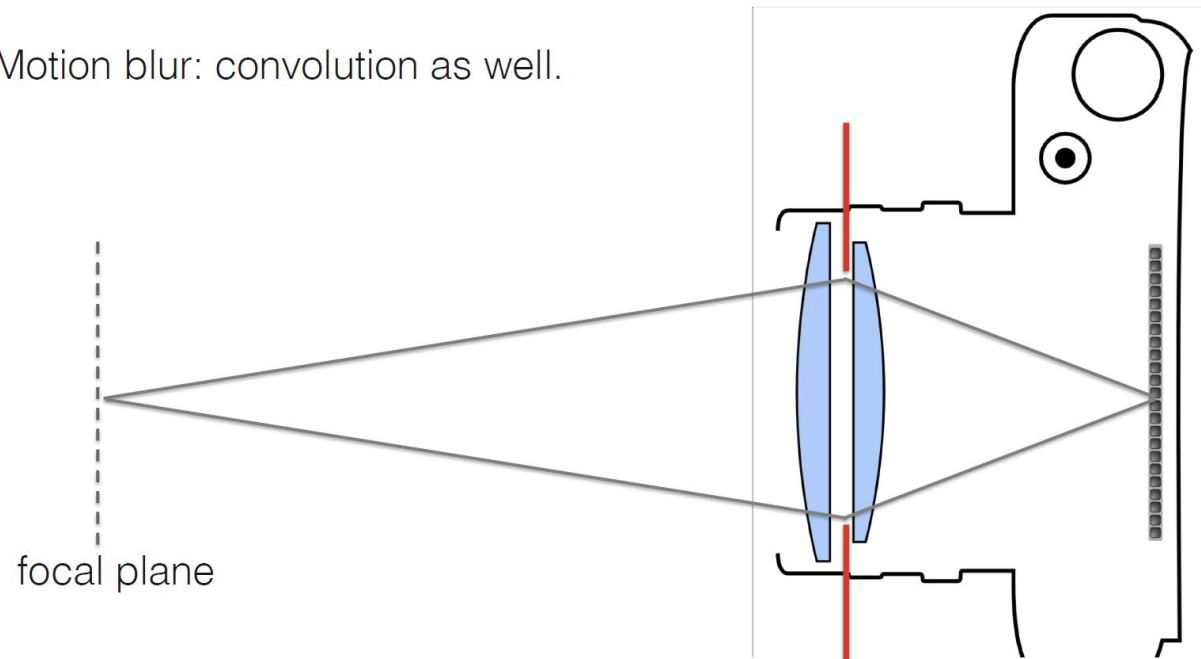
Defocus blur



$$g(x, y) = f(x, y) * h(x, y)$$

$$b(x, y) = \begin{cases} \frac{1}{\pi R^2}, & x^2 + y^2 \leq R^2 \\ 0, & \text{otherwise} \end{cases}$$

Motion blur: convolution as well.



During exposure, a moving point is recorded as a line:

point $\rightarrow$ line segment

For uniform linear motion, the PSF can be modeled as

$$h(x, y) = \begin{cases} \frac{1}{L}, & \text{along the motion path} \\ 0, & \text{otherwise} \end{cases}$$

Thus,

$$g(x, y) = f(x, y) * h_{\text{motion}}(x, y)$$

Motion blur can therefore be modeled as convolution with a line-shaped PSF.

Blurry Images

- An example of motion blur

Original signal: $f[n] = [0, 0, 1, 0, 0]$

Assume a uniform motion of length $L = 3$

Motion PSF

$$h[n] = [1/3 \quad 1/3 \quad 1/3], \text{ where } \sum_n h[n] = 1$$

Convolution

$$g[n] = f[n] * h[n]$$

$$[0, 0, 1, 0, 0] \rightarrow [0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0]$$

Original signal: $f[n] = [0, 1, 0, 2, 0]$

Assume a uniform motion of length $L = 3$

Motion PSF

$$h[n] = [1/3 \quad 1/3 \quad 1/3], \text{ where } \sum_n h[n] = 1$$

Convolution

$$g[n] = f[n] * h[n]$$

Contribution from the first point with intensity 1:

$$\left[0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0, 0, 0\right]$$

Contribution from the second point with intensity 2:

$$\left[0, 0, 0, \frac{2}{3}, \frac{2}{3}, \frac{2}{3}, 0\right]$$

Adding the two contributions:

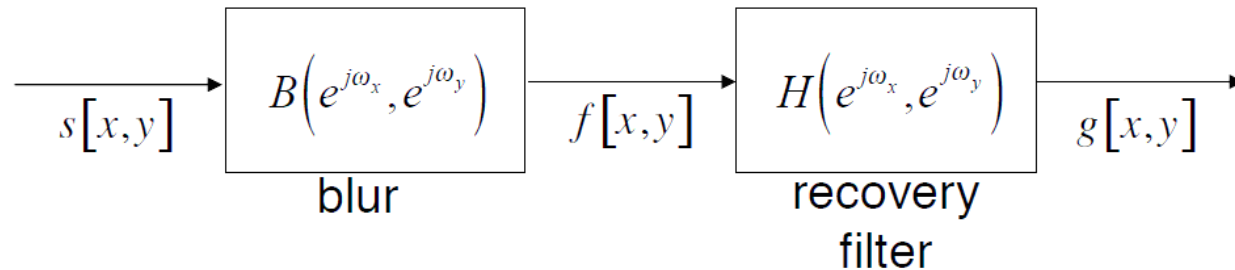
$$\left[0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3} + \frac{2}{3}, \frac{2}{3}, \frac{2}{3}, 0\right]$$

Therefore,

$$\boxed{\left[0, \frac{1}{3}, \frac{1}{3}, 1, \frac{2}{3}, \frac{2}{3}, 0\right]}$$

Image Deconvolution

- Given an image $f[x, y]$ that is a blurred version of original image $s[x, y]$, recover the original image
- Assume linear shift-invariant blur, transfer function $B(e^{j\omega_x}, e^{j\omega_y})$



- Naïve solution: inverse filter $H(e^{j\omega_x}, e^{j\omega_y}) = \frac{1}{B(e^{j\omega_x}, e^{j\omega_y})}$
- Problem: $B(e^{j\omega_x}, e^{j\omega_y})$ might be zero, noise amplification

Naïve Deconvolution

Original



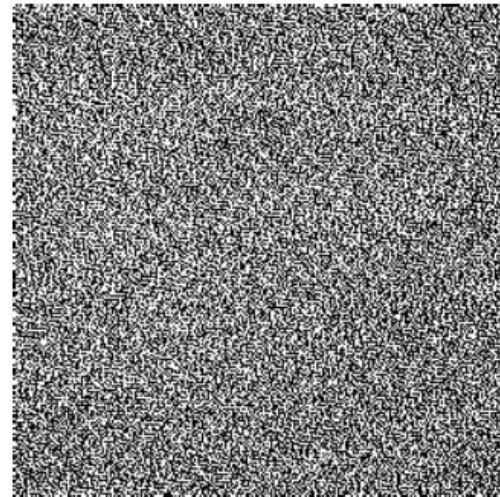
Blurred w/ $B(e^{j\omega_x}, e^{j\omega_y})$



Deblurred w/ $B^{-1}(e^{j\omega_x}, e^{j\omega_y})$



+ white noise $\downarrow$ $rms = 2.5$



$$rms = \sqrt{\frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N n^2(x, y)}$$

Blurr + Additive Noise

$$g(x, y) = f(x, y) * b(x, y) + n(x, y)$$

Using the convolution theorem,

$$f(x, y) * b(x, y) \iff F(\omega_x, \omega_y)B(\omega_x, \omega_y)$$

Therefore,

$$G(\omega_x, \omega_y) = B(\omega_x, \omega_y)F(\omega_x, \omega_y) + N(\omega_x, \omega_y)$$

For simplicity,

$$G = BF + N$$

If there is no noise,

$$G = BF$$

so

$$F = \frac{G}{B}$$

and the original image can ideally be recovered.

With noise,

$$G = BF + N$$

Applying the inverse filter gives

$$\begin{aligned}\hat{F} &= \frac{G}{B} \\ &= \frac{BF + N}{B}\end{aligned}$$

Therefore,

$$\hat{F} = F + \frac{N}{B}$$

Key Problem

When

$$|B(\omega_x, \omega_y)| \approx 0$$

then

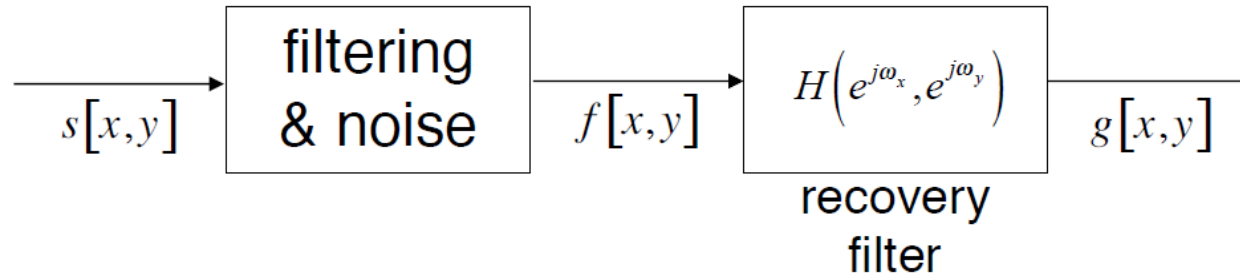
$$\left| \frac{N}{B} \right| \gg |N|$$

so even small noise can be strongly amplified.

Inverse filtering removes blur, but may strongly amplify noise where the blur transfer function is small.

Wiener Filtering

- Model



Power spectral density $\Phi_{ee} = F[E\{ee^*\}]$

- Minimize mean squared estimation error

$$E\{e^2[x, y]\} = E\{(g[x, y] - s[x, y])^2\} \xrightarrow{H(e^{j\omega_x}, e^{j\omega_y})} \min$$

- Power spectral density of estimator error ($(g - s)(g - s)^* = gg^* - gs^* - sg^* + ss^*$)

$$\begin{aligned} \Phi_{ee}(e^{j\omega_x}, e^{j\omega_y}) &= \Phi_{gg}(e^{j\omega_x}, e^{j\omega_y}) - \Phi_{gs}(e^{j\omega_x}, e^{j\omega_y}) - \Phi_{sg}(e^{j\omega_x}, e^{j\omega_y}) + \Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) \\ &= \Phi_{ff}(e^{j\omega_x}, e^{j\omega_y}) H(e^{j\omega_x}, e^{j\omega_y}) H^*(e^{j\omega_x}, e^{j\omega_y}) - \Phi_{fs}(e^{j\omega_x}, e^{j\omega_y}) H(e^{j\omega_x}, e^{j\omega_y}) \\ &\quad - \Phi_{sf}(e^{j\omega_x}, e^{j\omega_y}) H^*(e^{j\omega_x}, e^{j\omega_y}) + \Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) \end{aligned}$$

Cross-spectrum

Review: Power Spectrum and Cross Spectrum

2-d discrete-space cross correlation function for ergodic, stationary signals

$$\varphi_{fg}[m,n] = E \left\{ f[x+m, y+n] g^*[x, y] \right\}$$

Special case: autocorrelation function

$$\varphi_{ff}[m,n] = E \left\{ f[x+m, y+n] f^*[x, y] \right\}$$

Cross spectral density

$$\Phi_{fg}(e^{j\omega_x}, e^{j\omega_y}) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \varphi_{fg}[m,n] e^{-j\omega_x m - j\omega_y n}$$

Power spectral density

$$\Phi_{ff}(e^{j\omega_x}, e^{j\omega_y}) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \varphi_{ff}[m,n] e^{-j\omega_x m - j\omega_y n}$$

Wiener Filtering

- Power spectrum Φ_{ee} is minimized separately at each frequency ω_x, ω_y if

$$H(e^{j\omega_x}, e^{j\omega_y}) = \frac{\Phi_{fs}^*(e^{j\omega_x}, e^{j\omega_y})}{\Phi_{ff}(e^{j\omega_x}, e^{j\omega_y})} = \frac{\Phi_{sf}(e^{j\omega_x}, e^{j\omega_y})}{\Phi_{ff}(e^{j\omega_x}, e^{j\omega_y})}$$

Differentiate Φ_{ee} with respect to H

$$\frac{\partial \Phi_{ee}}{\partial H^*} = \Phi_{ff}H - \Phi_{sf}$$

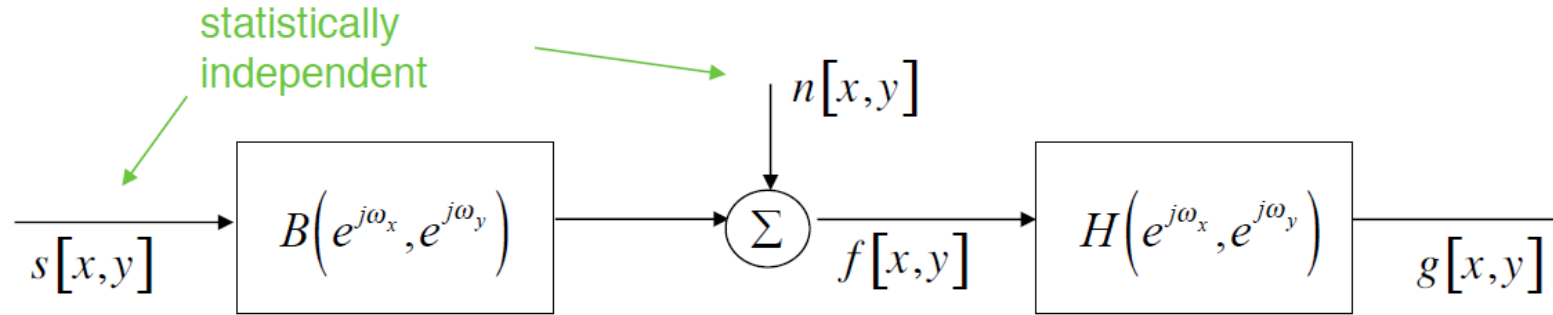
$$\Phi_{ff}H - \Phi_{sf} = 0$$

$$H = \frac{\Phi_{sf}}{\Phi_{ff}}$$

- Can be shown to be global minimum by considering filter

$$H(e^{j\omega_x}, e^{j\omega_y}) = \frac{\Phi_{fs}^*(e^{j\omega_x}, e^{j\omega_y})}{\Phi_{ff}(e^{j\omega_x}, e^{j\omega_y})} + \Delta H(e^{j\omega_x}, e^{j\omega_y})$$

Wiener Filter for Linear Distortion and Additive Noise



$$\Phi_{sf}(e^{j\omega_x}, e^{j\omega_y}) = \Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) B^*(e^{j\omega_x}, e^{j\omega_y})$$

$$\Phi_{ff}(e^{j\omega_x}, e^{j\omega_y}) = \Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) \left| B(e^{j\omega_x}, e^{j\omega_y}) \right|^2 + \Phi_{nn}(e^{j\omega_x}, e^{j\omega_y})$$

$$H(e^{j\omega_x}, e^{j\omega_y}) = \frac{\Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) B^*(e^{j\omega_x}, e^{j\omega_y})}{\Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) \left| B(e^{j\omega_x}, e^{j\omega_y}) \right|^2 + \Phi_{nn}(e^{j\omega_x}, e^{j\omega_y})}$$

Observation Model

$$F = BS + N$$

where

S : original image, B : linear distortion,

N : additive noise, F : observed image.

Assume that the signal and noise are statistically independent:

$$E\{SN^*\} = E\{NS^*\} = 0$$

Cross-Power Spectrum

$$\Phi_{sf} = E\{SF^*\}$$

Since

$$F^* = B^*S^* + N^*$$

we have

$$\begin{aligned}\Phi_{sf} &= E\{S(B^*S^* + N^*)\} \\ &= B^*E\{|S|^2\} + E\{SN^*\}\end{aligned}$$

Therefore,

$$\Phi_{sf} = \Phi_{ss}B^*$$

Power Spectrum of the Observed Image

$$\Phi_{ff} = E\{FF^*\}$$

Using

$$F = BS + N,$$

$$\begin{aligned}\Phi_{ff} &= E\{(BS + N)(B^*S^* + N^*)\} \\ &= |B|^2E\{|S|^2\} + BE\{SN^*\} + B^*E\{NS^*\} + E\{|N|^2\}.\end{aligned}$$

Since S and N are independent,

$$\Phi_{ff} = \Phi_{ss}|B|^2 + \Phi_{nn}$$

Wiener Filter

The optimal Wiener filter is

$$H = \frac{\Phi_{sf}}{\Phi_{ff}}.$$

Substituting the above results,

$$H = \frac{\Phi_{ss}B^*}{\Phi_{ss}|B|^2 + \Phi_{nn}}$$

or equivalently,

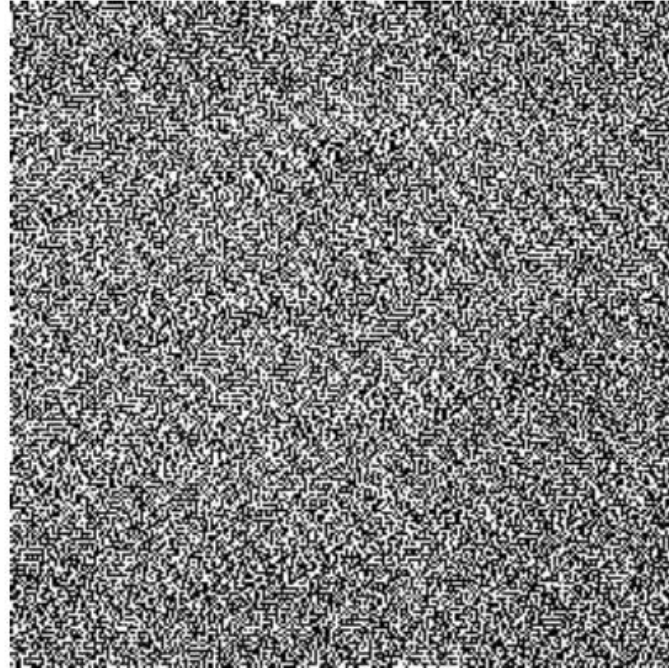
$$H = \frac{B^*}{|B|^2 + \frac{\Phi_{nn}}{\Phi_{ss}}}$$

Wiener Filter for Linear Distortion and Additive Noise

Blurred w/ $B(e^{j\omega_x}, e^{j\omega_y})$
Add. white noise, $rms = 2.5$

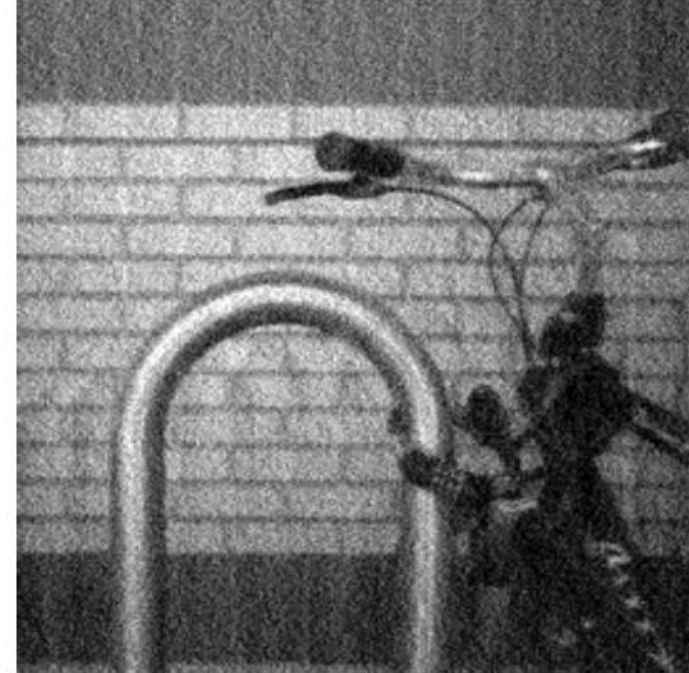


Deblurred
 $B^{-1}(e^{j\omega_x}, e^{j\omega_y})$

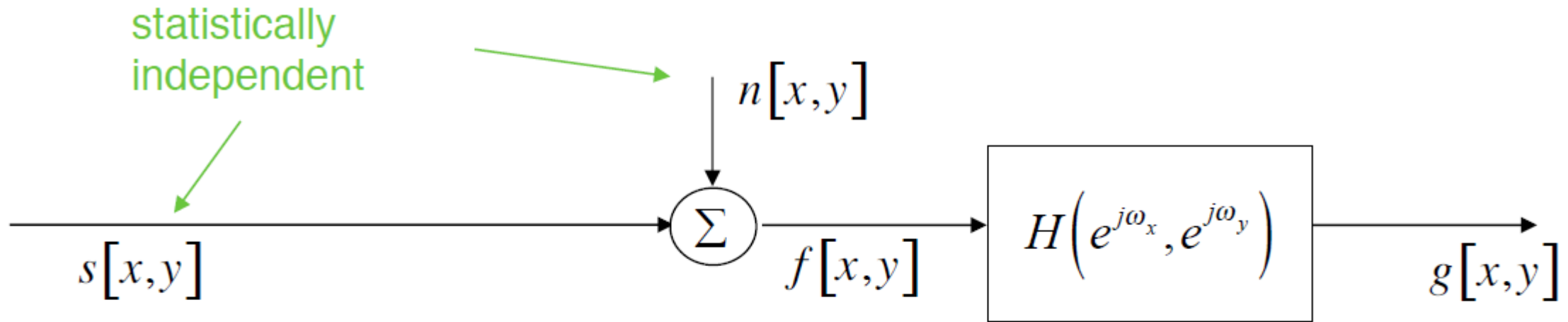


Wiener Filter

$$\frac{\Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) B^*(e^{j\omega_x}, e^{j\omega_y})}{\Phi_{ss}(e^{j\omega_x}, e^{j\omega_y}) \|B(e^{j\omega_x}, e^{j\omega_y})\|^2 + \Phi_{nn}(e^{j\omega_x}, e^{j\omega_y})}$$



Wiener Filter for Linear Distortion and Additive Noise



$$H\left(e^{j\omega_x}, e^{j\omega_y}\right) = \frac{\Phi_{ss}\left(e^{j\omega_x}, e^{j\omega_y}\right)}{\Phi_{ss}\left(e^{j\omega_x}, e^{j\omega_y}\right) + \Phi_{nn}\left(e^{j\omega_x}, e^{j\omega_y}\right)}$$

Wiener Filter for Additive Noise

Original



Additive
white noise



$$rms = 17.7$$

Noise reduction
by Wiener filtering

$$rms = 8.8$$



$$rms = \sqrt{\frac{1}{MN} \sum_{x=1}^M \sum_{y=1}^N (s(x,y) - \hat{s}(x,y))^2}$$

Wiener Filter: Design Perspective

$$H(\omega_x, \omega_y) = \frac{B^*(\omega_x, \omega_y)}{|B(\omega_x, \omega_y)|^2 + \frac{\Phi_{nn}(\omega_x, \omega_y)}{\Phi_{ss}(\omega_x, \omega_y)}}$$

1. B^* : Phase Compensation

$$B = |B|e^{j\theta_B} \Rightarrow B^* = |B|e^{-j\theta_B}$$

- Compensates for the phase distortion introduced by the blur.
- Provides the direction for inverse restoration.

$$B^* \rightarrow \text{Phase correction}$$

2. $|B|^2$: Inverse Filtering

If noise is negligible,

$$\frac{\Phi_{nn}}{\Phi_{ss}} \approx 0$$

then

$$H \approx \frac{B^*}{|B|^2} = \frac{1}{B}$$

Thus, the Wiener filter approaches the inverse filter.

$$\text{High SNR} \rightarrow \text{Strong deblurring}$$

Wiener Filter: Design Perspective

3. Φ_{nn}/Φ_{ss} : Noise Regularization

$$\frac{\Phi_{nn}}{\Phi_{ss}} = \text{Noise-to-Signal Power Ratio}$$

- Large ratio $\rightarrow$ noise dominates $\rightarrow$ reduce filter gain.
- Small ratio $\rightarrow$ signal is reliable $\rightarrow$ perform stronger inverse filtering.

Low SNR $\rightarrow$ Suppress restoration

In practice, we commonly approximate it as

$$\frac{\Phi_{nn}}{\Phi_{ss}} \approx K$$

and use Wiener filter in the form

$$H = \frac{B^*}{|B|^2 + K}$$

- Small $K \rightarrow$ stronger deblurring, but more noise amplification
- Large $K \rightarrow$ stronger noise suppression, but weaker deblurring

When

$$|B| \rightarrow 0$$

the inverse filter gives

$$\frac{1}{B} \rightarrow \infty$$

causing severe noise amplification.

However, the Wiener filter gives

$$H_W = \frac{B^*}{|B|^2 + \Phi_{nn}/\Phi_{ss}} \rightarrow 0$$

so unstable frequency components are suppressed.

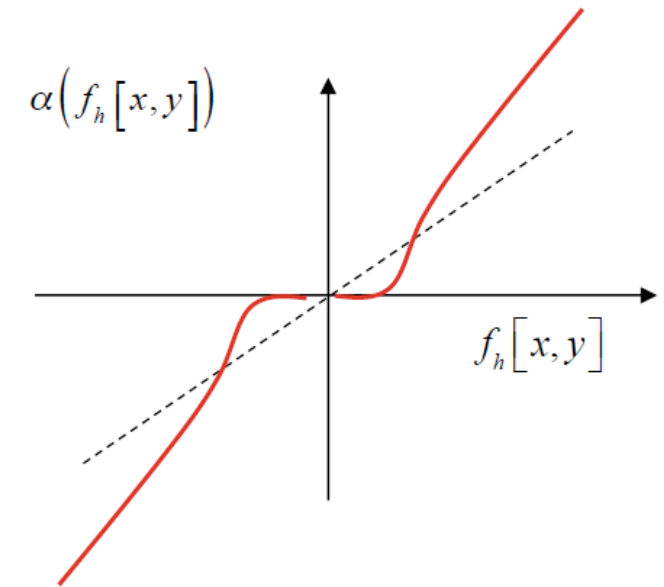
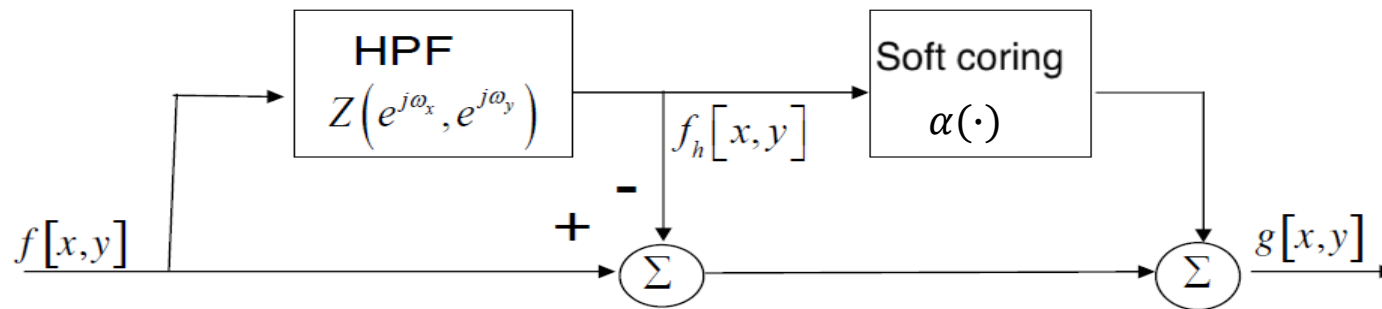
Wiener Filter: Design Perspective

$$\underbrace{B^*}_{\text{Phase correction}} / \left(\underbrace{|B|^2}_{\text{Deblurring}} + \underbrace{\Phi_{nn}/\Phi_{ss}}_{\text{Noise regularization}} \right)$$

The Wiener filter performs inverse filtering where the signal is reliable, while suppressing frequencies dominated by noise.

Nonlinear Noise Reduction/Sharpening

- Noise reduction: smooth the image, lowpass filtering
- Deblurring: sharpen edges, highpass filtering
- How can both be achieved simultaneously?
- Key insight: large amplitude of highpass filtered image indicates presence of edge



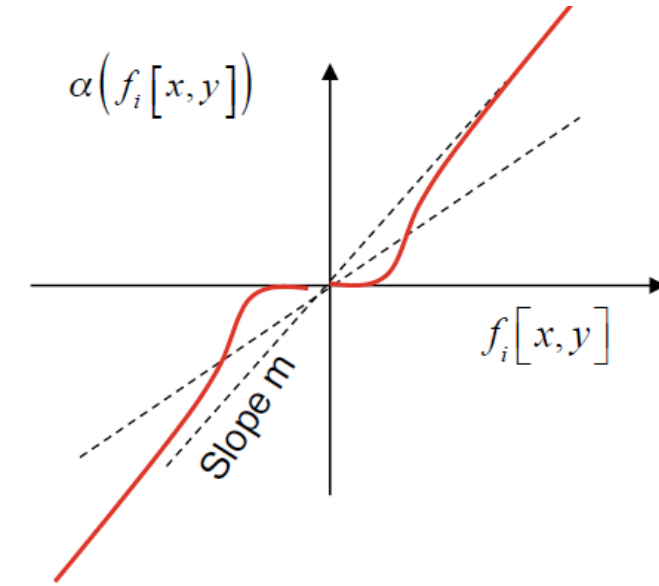
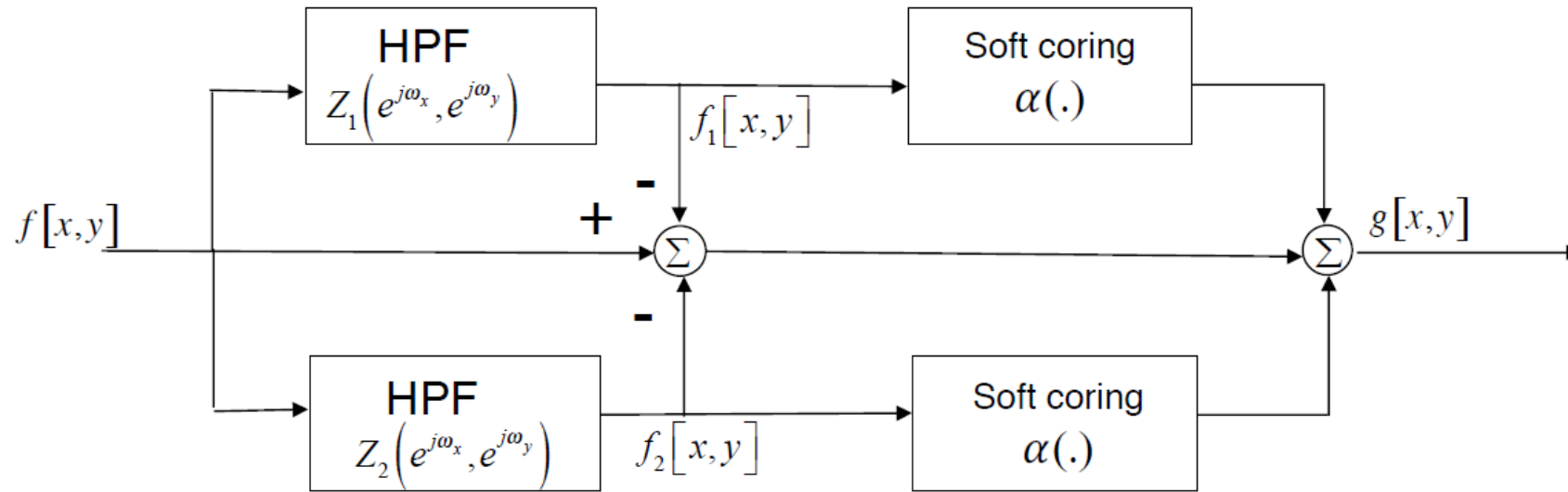
- Can be extended to multiple HPFs

$$g(x, y) = f_l[x, y] + \alpha(f_h(x, y))$$

$$\text{Using } f_l = f - f_h$$

$$g = f + [\alpha(f_h) - f_h]$$

Nonlinear Noise Reduction/Sharpening



Flat areas: $\alpha \approx 0 \rightarrow H(e^{j\omega_x}, e^{j\omega_y}) \approx 1 - Z_1(e^{j\omega_x}, e^{j\omega_y}) - Z_2(e^{j\omega_x}, e^{j\omega_y})$

f_1 small; f_2 large: $H(e^{j\omega_x}, e^{j\omega_y}) \approx 1 - Z_1(e^{j\omega_x}, e^{j\omega_y}) + (m-1)Z_2(e^{j\omega_x}, e^{j\omega_y})$

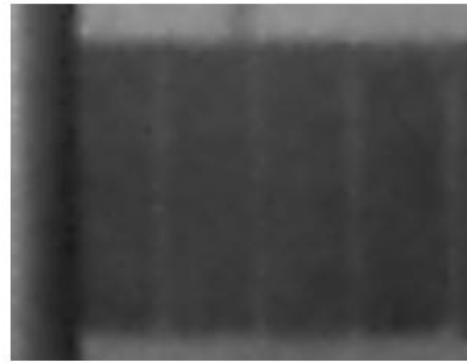
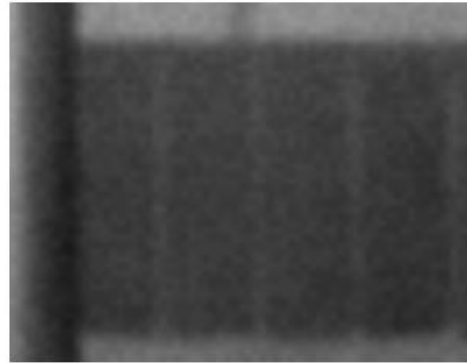
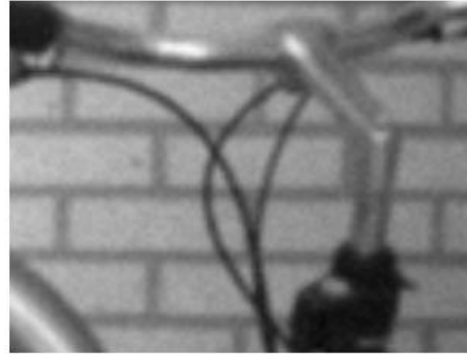
f_1 large; f_2 small: $H(e^{j\omega_x}, e^{j\omega_y}) \approx 1 + (m-1)Z_1(e^{j\omega_x}, e^{j\omega_y}) - Z_2(e^{j\omega_x}, e^{j\omega_y})$

Both large: $H(e^{j\omega_x}, e^{j\omega_y}) \approx 1 + (m-1)[Z_1(e^{j\omega_x}, e^{j\omega_y}) + Z_2(e^{j\omega_x}, e^{j\omega_y})]$

Nonlinear Noise Reduction/Sharpening



blurred, noisy image



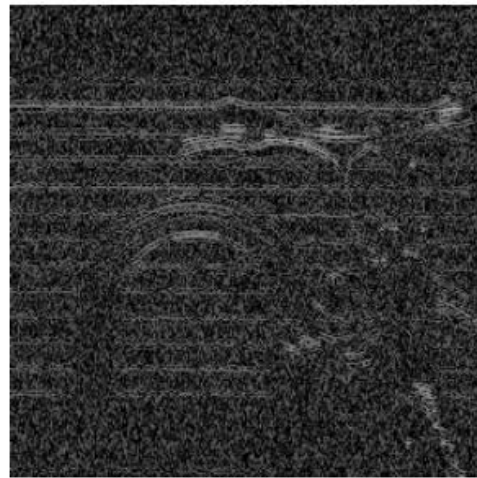
noise-reduced
and sharpened

Highpass Filtered Images



log magnitude of
image filtered with

$$\begin{pmatrix} 0 & 0 & 0 \\ -0.5 & [1] & -0.5 \\ 0 & 0 & 0 \end{pmatrix}$$



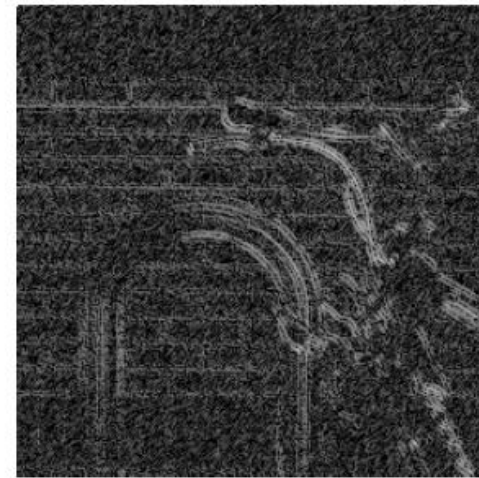
log magnitude of
image filtered with

$$\begin{pmatrix} 0 & -0.5 & 0 \\ 0 & [1] & 0 \\ 0 & -0.5 & 0 \end{pmatrix}$$



log magnitude of
image filtered with

$$\begin{pmatrix} -0.5 & 0 & 0 \\ 0 & [1] & 0 \\ 0 & 0 & -0.5 \end{pmatrix}$$

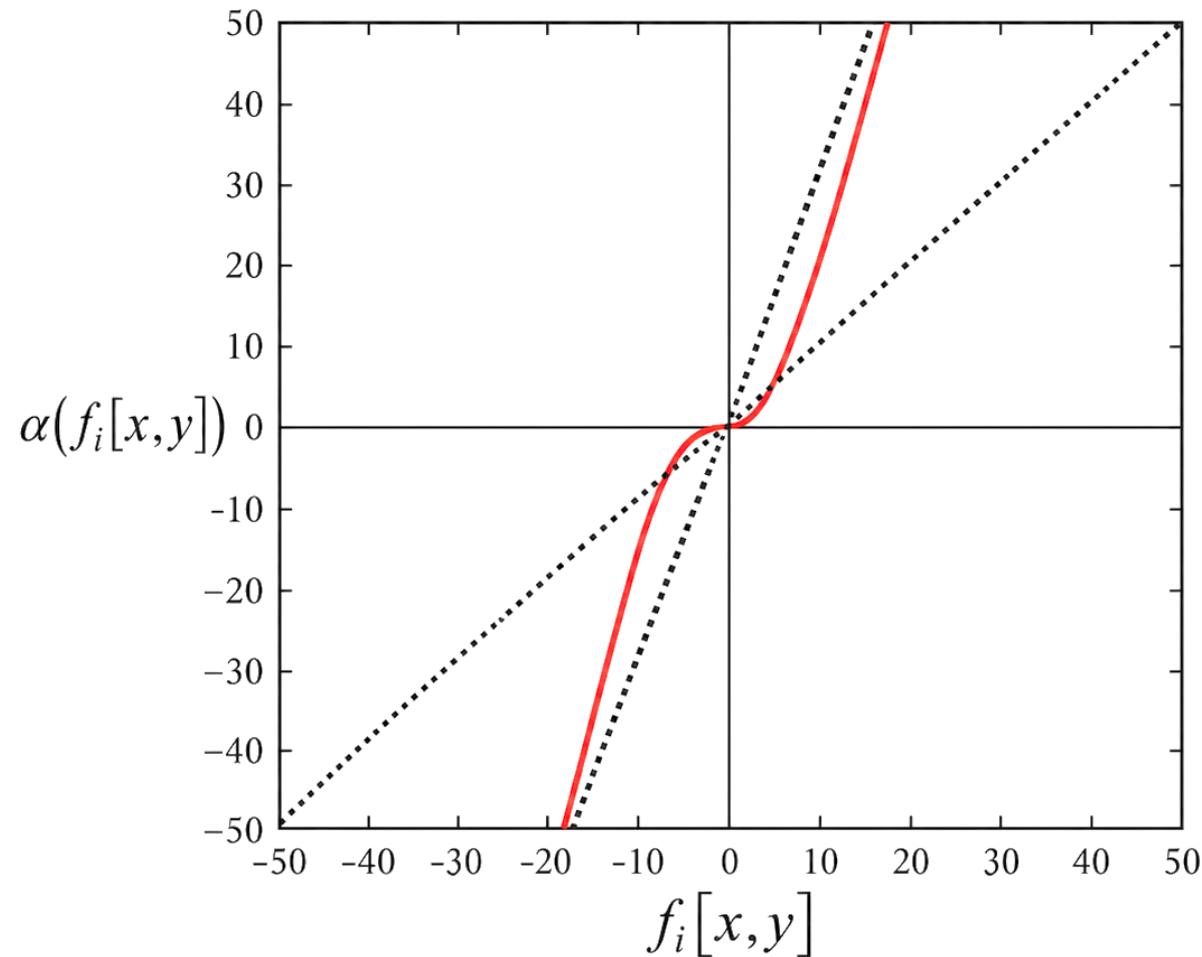


log magnitude of
image filtered with

$$\begin{pmatrix} 0 & 0 & -0.5 \\ 0 & [1] & 0 \\ -0.5 & 0 & 0 \end{pmatrix}$$

Soft Coring Function

$$\alpha(f_i[x,y]) = m \cdot f_i[x,y] \cdot \left(1 - e^{-\left| \frac{f_i[x,y]}{\tau} \right|^\gamma} \right)$$



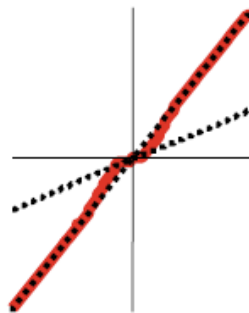
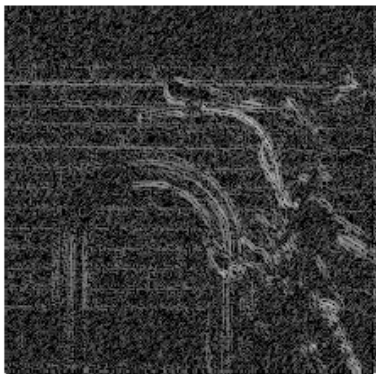
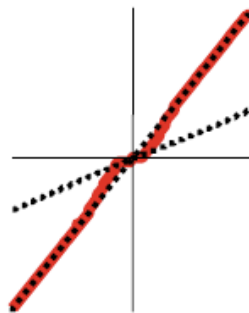
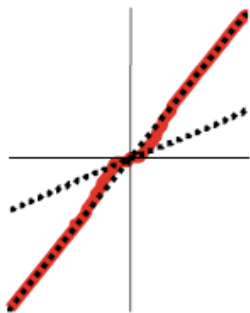
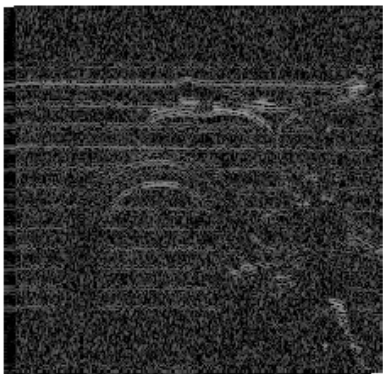
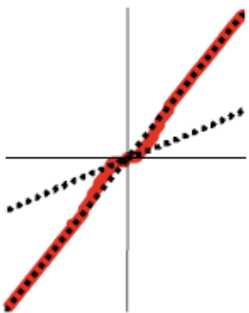
Example:

$$m = 3$$

$$\gamma = 2$$

$$\tau = 10$$

Soft Coring of Highpass Filtered Images



Linear vs. Nonlinear Noise Reduction/Sharpening



Noise reduction by
lowpass filter (linear)



Sharpening by highpass
filter (linear)



Combined noise reduction and
sharpening (nonlinear)